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PURE MATHEMATICS WITH STATISTICS • UPPER SIXTH

UPPER SIXTH — PURE MATHEMATICS WITH STATISTICS

CHAPTER 5

FIRST ORDER DIFFERENTIAL EQUATIONS WITH SEPARABLE VARIABLES

SUGGESTED TIME: 6 PERIODS

IN THIS CHAPTER

This chapter introduces first-order differential equations with separable variables, guiding students from the basic definition to real-life applications. Mastery of separation, integration, and initial-condition techniques is essential for success in the GCE Advanced Level Pure Mathematics with Statistics examination.

- Define a first-order differential equation and identify its order and degree.
- Recognise when a differential equation is separable.
- Apply the method of separation of variables to rewrite the equation in integrable form.
- Integrate both sides correctly to obtain the general solution, including the constant of integration.
- Use given initial conditions to determine the particular solution.
- Interpret the solution in the context of a real-world problem (population growth, cooling, chemical reactions, etc.).

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SUMMARY SHEET the essentials to remember

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CHAPTER OBJECTIVES

- Define a first-order differential equation and identify its order and degree.
- Recognise when a differential equation is separable.
- Apply the method of separation of variables to rewrite the equation in integrable form.
- Integrate both sides correctly to obtain the general solution, including the constant of integration.
- Use given initial conditions to determine the particular solution.
- Interpret the solution in the context of a real-world problem (population growth, cooling, chemical reactions, etc.).
- Check the validity of a solution by differentiation and by verifying the initial condition.
- Identify common algebraic and integration pitfalls and avoid them in exam answers.

PREREQUISITES

- Basic differentiation rules (power, exponential, trigonometric, logarithmic).
- Indefinite integration techniques (substitution, partial fractions, standard integrals).
- Algebraic manipulation of fractions and logarithms.
- Understanding of functions, domains, and implicit vs explicit forms.

- Experience with simple modelling problems from Lower Sixth (e.g., exponential growth/decay).

1 MEANING AND CLASSIFICATION OF FIRST ORDER DIFFERENTIAL EQUATIONS

Imagine a cocoa farmer in the South-West region who wants to predict how fast his cocoa pods will ripen depending on temperature and humidity. The rate of change of the ripening process can be modeled by a first-order differential equation. By learning to separate variables and integrate, you will be able to give the farmer a precise formula for the harvest time.

What is the farmer really asking? He is not asking how ripe the pods are today; he is asking how fast ripeness changes with time. In mathematics, a quantity whose **rate of change** is known — but whose value is not — is described by an equation involving a derivative. Such an equation is called a **differential equation**. This whole chapter is devoted to the simplest and most useful family of them: first-order equations whose variables can be separated. But before solving anything, we must understand exactly what these objects are.

1.1 What is a differential equation?

From a concrete situation to a definition.

Suppose a biologist observes that a colony of bacteria in a laboratory in Yaoundé grows at a rate proportional to its size. If $N(t)$ denotes the number of bacteria at time t (in hours), the observation translates into

$$\frac{dN}{dt} = kN,$$

where k is a positive constant. This is not an ordinary algebraic equation: the unknown is not a number but a function $N(t)$, and the equation links the function to its own derivative. Solving it means finding all functions whose derivative equals k times themselves.

DEFINITION **Differential equation**

A **differential equation** is an equation that relates an unknown function, its independent variable(s), and one or more of its derivatives. A solution of the equation is any function which, together with its derivatives, satisfies the equation identically on some interval.

EXEMPLE **Checking that a function is a solution**

Consider the equation $\frac{dy}{dx} = 2x$. The function $y = x^2 + 5$ is a solution, because $\frac{dy}{dx} = 2x$ for every real x. But so are $y = x^2 - 3$ and $y = x^2 + \sqrt{2}$. In fact every function $y = x^2 + C$, where C is an arbitrary constant, is a solution. This simple observation is fundamental: a differential equation usually has infinitely many solutions, indexed by constants of integration.

1.2 Ordinary versus partial differential equations

DEFINITION **ODE and PDE**

A differential equation is called **ordinary** (an ODE) when the unknown function depends on a single independent variable, so that only ordinary derivatives $\frac{dy}{dx}, \frac{d^2y}{dx^2}, \ldots$ appear. It is called **partial** (a PDE) when the unknown function depends on several independent variables and partial derivatives such as $\frac{\partial u}{\partial x}$ appear.

For example, $\frac{dy}{dx} = x - y$ is an ODE (one variable x), whereas the heat equation $\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}$, which describes how temperature $u(x, t)$ spreads along a metal bar, is a PDE (two variables x and t). **In this course we study only ordinary differential equations.**

1.3 Order and degree of a differential equation

Two numbers classify an ODE at a glance.

DEFINITION **Order and degree**

The **order** of a differential equation is the order of the highest derivative appearing in it. The **degree** of a differential equation is the power to which the highest derivative is raised, after the equation has been cleared of fractions and radicals involving the derivatives.

EXEMPLE **Finding order and degree**

1. $\frac{dy}{dx} + 3y = \sin x$: highest derivative is the first, raised to power 1. Order 1, degree 1.
2. $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + y = 0$: order 2, degree 1.
3. $\left(\frac{dy}{dx}\right)^3 + y = x$: order 1, degree 3.
4. $\sqrt{\frac{d^2y}{dx^2}} = y$: squaring gives $\frac{d^2y}{dx^2} = y^2$, so order 2, degree 1.
5. $\frac{d^3y}{dx^3} + \left(\frac{d^2y}{dx^2}\right)^2 = x$: order 3, degree 1 (the highest derivative is $\frac{d^3y}{dx^3}$ to the first power).

WARNING **Order is not degree**

Do not confuse the **order** (which derivative is the highest) with the **degree** (the power of that highest derivative after clearing fractions and roots). In $\left(\frac{d^2y}{dx^2}\right)^5 + \frac{dy}{dx} = 0$, the order is 2 and the degree is 5 — not the other way round. Examiners regularly test this distinction with a one-mark question.

DEFINITION **First-order ordinary differential equation**

A **first-order ordinary differential equation** is an equation involving an independent variable x, an unknown function $y(x)$ and its first derivative $\frac{dy}{dx}$ (and no higher derivative). It can always be written in the form

$$F\left(x,\, y,\, \frac{dy}{dx}\right) = 0,$$

and, when we can solve for the derivative, in the normal form

$$\frac{dy}{dx} = f(x, y).$$

The normal form $\frac{dy}{dx} = f(x, y)$ is the one we shall use throughout this chapter. It says: the slope of the unknown curve at the point (x, y) is a known number $f(x, y)$.

1.4 The concept of a solution: explicit and implicit

PROPERTY **What it means to be a solution**

A function φ is a solution of $\frac{dy}{dx} = f(x, y)$ on an interval I if φ is differentiable on I and, for every $x \in I$,

$$\varphi'(x) = f\big(x, \varphi(x)\big).$$

In words: substituting the function and its derivative into the equation produces a true statement at every point of the interval. (This characterisation is used constantly in examinations; its statement is examinable.)

Solutions come in two presentations.

DEFINITION **Explicit and implicit solutions**

A solution is **explicit** when it is given in the form $y = \varphi(x)$, with y isolated. It is **implicit** when it is given by a relation $G(x, y) = C$ which defines y as a function of x without y being isolated.

WORKED EXAMPLE 1 **Verifying an implicit solution**

Show that the relation $x^2 + y^2 = C$ (with $C > 0$) defines implicit solutions of the differential equation $x + y\dfrac{dy}{dx} = 0$.

SOLUTION

Solution. Differentiate both sides of $x^2 + y^2 = C$ with respect to x, remembering that y is a function of x (chain rule: $\dfrac{d}{dx}(y^2) = 2y\dfrac{dy}{dx}$):

$$2x + 2y\frac{dy}{dx} = 0 \quad \Longrightarrow \quad x + y\frac{dy}{dx} = 0.$$

This is exactly the given equation, and it holds wherever $y \neq 0$ (so that the circle has a well-defined tangent slope). Hence $x^2 + y^2 = C$ is an implicit solution. Geometrically, the solutions are the circles centred at the origin: at each point of a circle, the tangent is perpendicular to the radius, which is precisely what $x + y\,y' = 0$ expresses.

WARNING **A solution must be checked on an interval**

A formula is only a solution where it is differentiable and where the substitution makes sense. For instance $y = \sqrt{C - x^2}$ solves $x + yy' = 0$ only on $(-\sqrt{C}, \sqrt{C})$, not at the endpoints where the tangent is vertical. Always state the interval when asked to verify a solution.

1.5 Where first-order equations come from

First-order ODEs appear whenever a rate of change depends on the present state. Here are three classical sources, all of which we shall meet again in the applications section.

Physics: Newton's law of cooling.

A bottle of water taken from a fridge in Douala warms up at a rate proportional to the difference between its temperature $T(t)$ and the ambient temperature T_a:

$$\frac{dT}{dt} = -k\,(T - T_a), \qquad k > 0.$$

Biology: limited (logistic) growth.

When a population $P(t)$ cannot exceed a carrying capacity K (think of fish in a pond near Limbe),

a refined model is

$$\frac{dP}{dt} = rP\left(1 - \frac{P}{K}\right).$$

Economics: continuous compound interest.

If a savings account at a credit union earns interest continuously at rate r, the balance $M(t)$ satisfies

$$\frac{dM}{dt} = rM.$$

Notice the common pattern: in each case nature gives us the slope, and mathematics must recover the curve.

1.6 Graphical interpretation: slope fields and integral curves

Since $\frac{dy}{dx} = f(x, y)$ tells us the slope of the solution curve passing through any point (x, y), we can see all the solutions at once without solving anything: at each point of a grid, draw a short segment with slope $f(x, y)$. The resulting picture is called a **slope field** (or direction field). A curve that follows the little segments everywhere is called an **integral curve** — it is the graph of a solution.

METHOD **Sketching a slope field by hand**

1. Choose a grid of points, for example $x, y \in \{-2, -1, 0, 1, 2\}$.
2. At each grid point (x, y), compute the slope $m = f(x, y)$. Organise the values in a table of slopes (rows: values of y; columns: values of x).
3. At each point, draw a short line segment with the computed slope (horizontal if $m = 0$, steep if $|m|$ is large).
4. To draw an integral curve, start at the required point and sketch a smooth curve that stays tangent to the segments.

EXEMPLE **Table of slopes for $\frac{dy}{dx} = x - y$**

The slope depends only on $x - y$, so it is constant along each line $x - y = c$ (such lines are called isoclines).

$y \backslash x$	-2	-1	0	1	2
2	-4	-3	-2	-1	0
1	-3	-2	-1	0	1
0	-2	-1	0	1	2
-1	-1	0	1	2	3
-2	0	1	2	3	4

Along the line $y = x$ every slope is 0: solutions cross this line horizontally. Below it slopes are positive, above it negative.



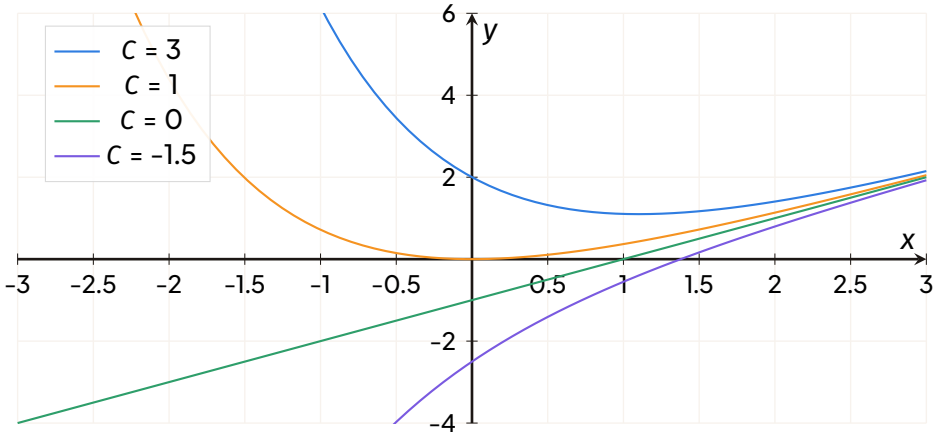
The equation $\frac{dy}{dx} = x - y$ can in fact be solved exactly (it is linear; you will meet such equations later). Its solutions are

$$y = Ce^{-x} + x - 1, \qquad C \in \mathbb{R}.$$

Let us verify this claim directly, as the property above requires: if $y = Ce^{-x}+x-1$, then $\frac{dy}{dx} = -Ce^{-x}+1$, and

$$x - y = x - (Ce^{-x} + x - 1) = -Ce^{-x} + 1 = \frac{dy}{dx}.$$

The equation is satisfied for every real x, so $y = Ce^{-x}+x-1$ is indeed the **general solution**: a family of solutions depending on one arbitrary constant C. Each value of C gives one integral curve of the slope field above.



Explicit solutions $y = Ce^{-x} + x - 1$ for several values of C. For large x the term Ce^{-x} dies out and every curve approaches the line $y = x - 1$.

KEY POINTS **Key points of this section**

- A differential equation links an unknown function to its derivatives; solving it means finding functions.
- The **order** is the highest derivative present; the **degree** is the power of that derivative after clearing fractions and radicals.
- A first-order ODE in normal form reads $\frac{dy}{dx} = f(x, y)$: the slope of the solution is prescribed at every point.
- A solution may be given explicitly ($y = \varphi(x)$) or implicitly ($G(x, y) = C$); it must satisfy the equation on an interval where it is differentiable.
- The general solution of a first-order ODE contains one arbitrary constant; geometrically it is a one-parameter family of integral curves of the slope field.

DID YOU KNOW? **Why one constant?**

A first-order equation involves one derivative, and undoing one derivative requires one integration — hence one constant of integration. Likewise, a second-order equation (like those governing the motion of a pendulum or the vibration of a balafon string) produces two constants. This is why the general solution of a first-order ODE is always a family of curves, and why one extra piece of information — an initial condition — will be needed to select a single curve, as we shall see in Section 4.

EXERCISE 1 **Classifying differential equations**

For each equation, give the order and the degree, and state whether it is ordinary or partial.

1. $\frac{dy}{dx} = \frac{x^2}{y}$
2. $\left(\frac{d^2y}{dx^2}\right)^2 + \left(\frac{dy}{dx}\right)^3 = y$
3. $\frac{\partial u}{\partial t} = 3\frac{\partial u}{\partial x}$
4. $1 + \left(\frac{dy}{dx}\right)^2 = \frac{1}{y^2}$

SOLUTION **Exercise 1**

1. Highest derivative: $\frac{dy}{dx}$, power 1. Order 1, degree 1; ordinary.
2. Highest derivative: $\frac{d^2y}{dx^2}$, raised to the power 2. Order 2, degree 2; ordinary.
3. Partial derivatives appear: it is a PDE, of order 1 and degree 1.
4. Highest derivative: $\frac{dy}{dx}$, power 2 (no fractions in the derivatives to clear). Order 1, degree 2; ordinary.

EXERCISE 2 **Verifying a solution**

Show that $y = \frac{1}{1-x}$ is a solution of $\frac{dy}{dx} = y^2$, and state the largest interval containing $x = 0$

on which this solution is valid.

SOLUTION **Exercise 2**

If $y = (1 - x)^{-1}$, then $\frac{dy}{dx} = (1 - x)^{-2} = y^2$, so the equation is satisfied. The function is differentiable everywhere except at $x = 1$ where it is undefined. The largest interval containing 0 on which it is a solution is therefore $(-\infty, 1)$.

Now that we can recognise, classify and visualise first-order differential equations, the next step is to identify those that can be solved by the simplest and most powerful technique of this chapter: separation of variables. That is the object of the next section.

2 RECOGNISING SEPARABLE EQUATIONS

In Section 1 we learned to recognise a first-order differential equation: an equation involving y, x and $\frac{dy}{dx}$, and we classified such equations by order and degree. We also saw that there is no universal formula that solves every first-order equation — each type of equation has its own method. The very first question a GCE candidate must therefore ask when facing a differential equation is: **which type is it?** In this section we study the simplest and most frequently examined type at Advanced Level: equations whose variables can be **separated**. By the end of the section you will be able to decide, quickly and with justification, whether a given equation is separable, and to carry out the algebraic manipulation that puts it into separated form, ready for the integration of Section 3.

2.1 The Idea of Separation: an Intuitive Approach

Think of the equation $\frac{dy}{dx} = 2xy$. The right-hand side is a product: a piece that depends only on x (namely $2x$) times a piece that depends only on y (namely y). Because the two variables are "unmixed", we can send everything involving y to the left with dy, and everything involving x to the right with dx:

$$\frac{1}{y}\,dy = 2x\,dx.$$

Each side now contains one single variable, and each side can be integrated on its own. This is the whole philosophy of separation of variables.

Contrast this with $\frac{dy}{dx} = x+y$. Here the right-hand side is a sum in which x and y are glued together by the + sign. No amount of division or factorisation will produce a product of an x-part and a y-part: the variables are genuinely mixed. This equation is not separable (it will be solved later by other methods).

DEFINITION **Separable differential equation**

A first-order differential equation is said to be **separable** (or to have **separable variables**) if it can be written in the form

$$\frac{dy}{dx} = H(x)\,G(y),$$

that is, the right-hand side is a product of a function of x alone and a function of y alone. Equivalently, it can be rearranged into the separated form

$$g(y)\,dy = h(x)\,dx,$$

where g depends only on y and h depends only on x.

WARNING **The criterion is a PRODUCT, not a sum**

Separation is possible precisely when x and y appear as factors on the right-hand side. Sums and differences such as $x + y$, $x^2 - y^2$ (which does not factor as a product of single-variable functions because $(x-y)(x+y)$ still mixes the variables), or e^{x+y} handled carelessly, are the classic traps. Note that $e^{x+y} = e^x e^y$ is a product, so $\frac{dy}{dx} = e^{x+y}$ is separable, whereas $\frac{dy}{dx} = x + y$ is not.

2.2 The Key Property Justifying Separation

PROPERTY **Separation of variables**

If $\frac{dy}{dx} = f(x)\,g(y)$ and $g(y) \neq 0$ on the interval considered, then the equation is equivalent to

$$\frac{1}{g(y)}\,dy = f(x)\,dx.$$

(The proof that integrating both sides yields the solution is examinable and will be given in Section 3; here we focus on the algebraic step.)

Why division by $g(y)$ is legitimate. On any interval where $g(y) \neq 0$, the function $\frac{1}{g(y)}$ is well defined, so we may multiply both sides of $\frac{dy}{dx} = f(x)g(y)$ by $\frac{dx}{g(y)}$. Formally, using the chain rule, $\frac{1}{g(y)}\frac{dy}{dx} = f(x)$ says that the antiderivative of $\frac{1}{g(y)}$ with respect to y and the antiderivative of $f(x)$ with respect to x have the same derivative with respect to x; hence they differ only by a constant. This is exactly what justifies integrating the two sides separately.

2.3 Algebraic Manipulation to Achieve Separation

Most examination equations are not handed to you already in product form: you must manufacture the product. The three standard tools are **factorisation**, **division** and **multiplication**. Let us see them at work.

EXEMPLE **Factorising to reveal the product**

Consider $\frac{dy}{dx} = xy + x$. At first sight the right-hand side is a sum. But factorising gives

$$xy + x = x(y + 1),$$

a product of a function of x and a function of y. Hence the equation is separable, and for $y \neq -1$:

$$\frac{dy}{y+1} = x\,dx.$$

EXEMPLE **Dividing both sides**

Consider $x\frac{dy}{dx} = y(1 + x^2)$. Divide both sides by x (for $x \neq 0$):

$$\frac{dy}{dx} = \frac{y(1+x^2)}{x} = y \cdot \frac{1+x^2}{x},$$

a product of a function of y and a function of x. Separating (for $y \neq 0$):

$$\frac{dy}{y} = \frac{1+x^2}{x}\,dx.$$

EXEMPLE **Cross-multiplication from a quotient form**

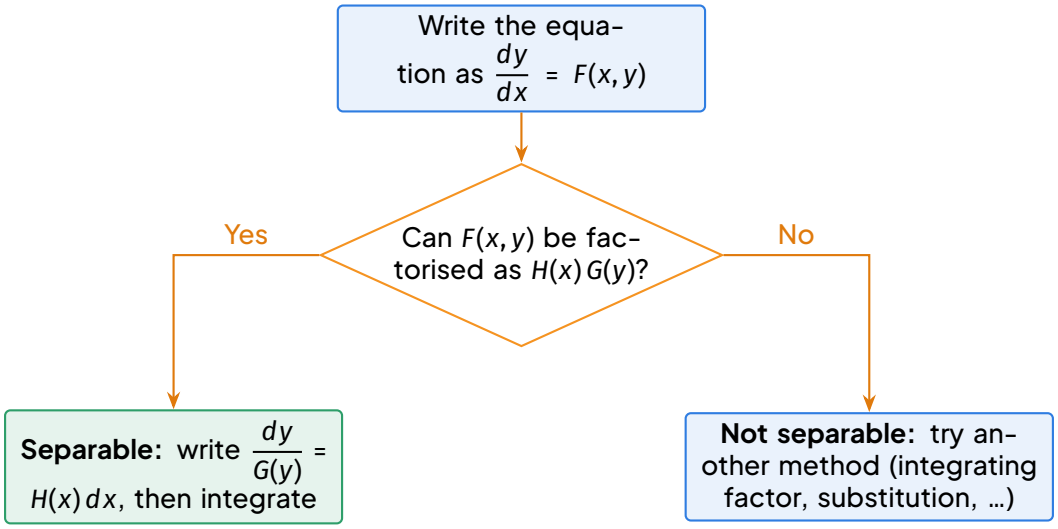
Consider $\frac{dy}{dx} = \frac{x\,e^x}{y^2}$. The right-hand side is already $H(x)G(y)$ with $H(x) = xe^x$ and $G(y) = \frac{1}{y^2}$. Multiplying through by $y^2\,dx$ gives the separated form directly:

$$y^2\,dy = xe^x\,dx.$$

METHOD **Checklist to test separability**

To decide whether $\frac{dy}{dx} = F(x, y)$ is separable:

1. Isolate $\frac{dy}{dx}$ on one side if necessary (clear brackets, divide by the coefficient of $\frac{dy}{dx}$).
2. Try to write the other side as a product $H(x)\,G(y)$: factorise, use index laws ($e^{x+y} = e^x e^y$), or split quotients.
3. If a product is obtained, the equation is separable: divide by $G(y)$ and multiply by dx to reach $g(y)\,dy = h(x)\,dx$.
4. If no factorisation is possible (typically because of a sum $x + y$ or a genuinely mixed expression such as $\sin(xy)$), the equation is not separable — do not force it.
5. Record any values excluded by your divisions (e.g. $y = 0$): they may be lost solutions.



Decision flowchart: testing whether a first-order equation is separable.

2.4 Recognising Non-Separable Forms

Knowing when separation fails is as valuable as knowing when it works: it saves precious examination time.

EXEMPLE **Equations that are NOT separable**

- $\frac{dy}{dx} = x + y$: a sum; no factorisation $H(x)G(y)$ exists.
- $\frac{dy}{dx} = x^2 + y^2$: a sum of squares; not factorisable into single-variable factors over $\mathbb{R}$.

- $\frac{dy}{dx} = \sin(x + y)$ or $\frac{dy}{dx} = e^{xy}$: the variables are mixed inside another function.
- $\frac{dy}{dx} = \frac{x + y}{x - y}$: this is a homogeneous equation (right-hand side depends only on $\frac{y}{x}$); it is solved by the substitution $v = \frac{y}{x}$, not by direct separation.

WARNING **Homogeneous does not mean separable**

A frequent confusion at GCE: an equation like $\frac{dy}{dx} = \frac{y}{x}$ is both homogeneous and separable (it equals $y \cdot \frac{1}{x}$), but $\frac{dy}{dx} = \frac{x + y}{x - y}$ is homogeneous without being separable. Test for the product form first; never assume.

WARNING **Lost solutions when dividing by $g(y)$**

When we divide by $g(y)$ we silently assume $g(y) \neq 0$. Any constant $y = c$ with $g(c) = 0$ is in fact a solution of the original equation (since then $\frac{dy}{dx} = 0 = f(x) \cdot 0$), but it is lost in the division. Example: separating $\frac{dy}{dx} = 2xy$ by dividing by y loses the solution $y = 0$. Always check whether the excluded constant solutions satisfy the equation and the initial condition.

2.5 Worked Example and Classification Practice

WORKED EXAMPLE 2 **Separating a disguised equation**

Show that $xy\,\frac{dy}{dx} = y^2 + 1$ is separable, and write it in the form $g(y)\,dy = h(x)\,dx$.

SOLUTION

Solution. Divide both sides by x (valid for $x \neq 0$):

$$\frac{dy}{dx} = \frac{y^2 + 1}{xy} = \frac{1}{x} \cdot \frac{y^2 + 1}{y}.$$

This is a product $H(x)G(y)$ with $H(x) = \frac{1}{x}$ and $G(y) = \frac{y^2 + 1}{y}$, so the equation is separable. Mul-

tiplying by $\frac{y}{y^2+1}\,dx$ (note $y^2+1 \neq 0$ for all real y, so no solution is lost):

$$\frac{y}{y^2+1}\,dy = \frac{1}{x}\,dx.$$

This is the required separated form, ready for integration.

EXERCISE 3 **Classify ten equations**

For each equation, state whether it is separable or not separable, and justify briefly. When separable, give the separated form.

1. $\frac{dy}{dx} = 3x^2y$
2. $\frac{dy}{dx} = x + 2y$
3. $\frac{dy}{dx} = e^{x-y}$
4. $\frac{dy}{dx} = \frac{x}{y} + \frac{y}{x}$
5. $(1+x^2)\frac{dy}{dx} = 1 + y^2$
6. $\frac{dy}{dx} = \sin x \cos y$
7. $\frac{dy}{dx} = \sin(xy)$
8. $y\frac{dy}{dx} = x(y^2 - 4)$
9. $\frac{dy}{dx} = \frac{x^2 + x}{y}$
10. $\frac{dy}{dx} = (x+y)^2$

SOLUTION **Exercise 1**

1. Separable: $3x^2y = (3x^2)(y)$. For $y \neq 0$: $\frac{dy}{y} = 3x^2\,dx$. (Check: $y = 0$ is also a solution.)
2. Not separable: $x + 2y$ is a sum and cannot be written as $H(x)G(y)$.

3. Separable: $e^{x-y} = e^x e^{-y}$, a product. $e^y\,dy = e^x\,dx$.
4. Not separable: $\frac{x}{y} + \frac{y}{x} = \frac{x^2+y^2}{xy}$ cannot be factored into single-variable functions (it is homogeneous).
5. Separable: $\frac{dy}{dx} = \frac{1+y^2}{1+x^2} = \frac{1}{1+x^2}(1+y^2)$. Hence $\frac{dy}{1+y^2} = \frac{dx}{1+x^2}$.
6. Separable: product $\sin x \cdot \cos y$. For $\cos y \neq 0$: $\frac{dy}{\cos y} = \sin x\,dx$.
7. Not separable: the variables are mixed inside the sine.
8. Separable: $\frac{dy}{dx} = \frac{x(y^2-4)}{y} = x \cdot \frac{y^2-4}{y}$. Hence $\frac{y}{y^2-4}\,dy = x\,dx$ (for $y \neq \pm 2$; check $y = \pm 2$ separately).
9. Separable: $\frac{dy}{dx} = (x^2+x) \cdot \frac{1}{y}$. Hence $y\,dy = (x^2+x)\,dx$.
10. Not separable: $(x+y)^2 = x^2 + 2xy + y^2$ contains the mixed term $2xy$; no product form exists (a substitution $u = x + y$ is needed).

2.6 A Brief Link with Exact Equations

DID YOU KNOW? **Separable implies exact — but not conversely**

An equation written as $M(x)\,dx + N(y)\,dy = 0$ is called **exact** when M and N are the partial derivatives of some potential function. Every separable equation $g(y)\,dy - h(x)\,dx = 0$ is automatically exact, because the "cross-derivatives" are both zero: $\frac{\partial}{\partial y}(-h(x)) = 0 = \frac{\partial}{\partial x}(g(y))$.

The converse is false: for example $2xy\,dx + x^2\,dy = 0$ is exact (it is the differential of x^2y) yet its separated structure is hidden. At GCE Advanced Level you only need the separable case, but it is reassuring to know that separation of variables sits inside a larger, elegant theory.

KEY POINTS **Key points of this section**

- An equation is **separable** if it can be written $\frac{dy}{dx} = H(x)G(y)$, equivalently $g(y)\,dy = h(x)\,dx$.
- The test is: can the right-hand side be factored into an x-part times a y-part? Use factorisation, division, multiplication and index laws.

- Sums ($x + y$), mixed arguments ($\sin(xy)$, e^{xy}) and most homogeneous equations are not separable.
- Dividing by $g(y)$ assumes $g(y) \neq 0$: always check the constant solutions $g(y) = 0$ that may have been lost.
- Once separated, each side contains a single variable — the stage is set for integration, which is the object of the next section.

3 SEPARATION OF VARIABLES AND INTEGRATION TO OBTAIN GENERAL SOLUTIONS

In Section 2 we learned to recognise a separable equation: one that can be written in the form

$$\frac{dy}{dx} = f(x)\,g(y).$$

Recognition, however, is only the passport. We now learn the journey itself: how to solve such an equation completely. The strategy is beautifully simple — put all the y's with dy on one side, all the x's with dx on the other, and integrate. The skill lies in doing this correctly: handling the constant of integration, the absolute values that appear in logarithms, and the final step of making y the subject when the question demands it. This section is the computational heart of the chapter, and it is where most GCE marks on differential equations are won or lost.

3.1 The Principle of Separation

Consider the separable equation $\frac{dy}{dx} = f(x)\,g(y)$, with $g(y) \neq 0$ on the interval of interest. Dividing both sides by $g(y)$ gives

$$\frac{1}{g(y)}\frac{dy}{dx} = f(x).$$

Now integrate both sides with respect to x:

$$\int \frac{1}{g(y)}\frac{dy}{dx}\,dx = \int f(x)\,dx.$$

By the rule of integration by substitution (with y as the new variable), the left-hand side is exactly $\int \frac{1}{g(y)}\,dy$. Hence the equation is equivalent to

$$\boxed{\int \frac{1}{g(y)}\,dy = \int f(x)\,dx + C}$$

This is the **formal separation** of variables. Informally, students often remember it as "multiply through by dx and divide by $g(y)$":

$$\frac{dy}{g(y)} = f(x)\,dx,$$

then attach integral signs. Although dy and dx are not ordinary numbers, this shorthand is justified rigorously by the substitution rule above, so you may use it confidently in the examination.

PROPERTY **One Constant of Integration Only**

When both sides of a separated equation are integrated, only one arbitrary constant is needed. If the left integral gives $G(y) + C_1$ and the right gives $F(x) + C_2$, then

$$G(y) + C_1 = F(x) + C_2 \quad \Longrightarrow \quad G(y) = F(x) + (C_2 - C_1).$$

Since $C_2 - C_1$ is itself just an arbitrary constant, we rename it C and write $G(y) = F(x) + C$. Writing two constants is not wrong, but it is untidy and wastes time; examiners expect a single constant C.

METHOD **The Separation–Integration Algorithm**

To solve a separable first-order differential equation $\dfrac{dy}{dx} = f(x)\,g(y)$:

1. Separate. Rearrange so that all terms in y multiply dy and all terms in x multiply dx: $\dfrac{dy}{g(y)} = f(x)\,dx$.
2. Integrate both sides. Write $\displaystyle\int \frac{dy}{g(y)} = \int f(x)\,dx$ and evaluate each integral using the appropriate technique (power rule, exponential, logarithmic, trigonometric, or partial fractions).
3. Insert one constant. Add a single arbitrary constant C on one side only.
4. Simplify. Handle absolute values (e.g. $\ln|y|$), combine logarithms, and exponentiate where useful.
5. Make y the subject if required. Give the general solution explicitly as $y = \varphi(x)$ when possible; otherwise leave it in implicit form.

3.2 Integration Techniques You Must Command

Separation reduces every problem in this chapter to two ordinary integrals. The GCE examiner will therefore test your integration fluency. The techniques required are:

Technique	Typical occurrence	Result
Power rule	$\int x^n\,dx$, $\int y^n\,dy$	$\frac{x^{n+1}}{n+1}$ $(n \neq -1)$
Logarithmic	$\int \frac{dy}{y}$, $\int \frac{f'(x)}{f(x)}\,dx$	$\ln\lvert y\rvert$, $\ln\lvert f(x)\rvert$
Exponential	$\int e^{kx}\,dx$	$\frac{1}{k}e^{kx}$
Trigonometric	$\int \sin x\,dx$, $\int \sec^2 y\,dy$	$-\cos x$, $\tan y$
Partial fractions	$\int \frac{dy}{y(a-y)}$	split, then logarithms

The last row deserves special attention: equations such as $\frac{dy}{dx} = ky(a - y)$ (the logistic-type equations that appear in population models) separate into an integral that only partial fractions can unlock. We shall meet this in Section 5.

3.3 The Absolute Value in $\ln\lvert y\rvert$: a Non-Negotiable Detail

When separation produces $\int \frac{dy}{y}$, the answer is $\ln\lvert y\rvert$, not $\ln y$. The reason is fundamental: the logarithm $\ln y$ is only defined for $y > 0$, whereas the solutions of a differential equation may well be negative. The antiderivative of $1/y$ that is valid on every interval not containing 0 is $\ln\lvert y\rvert$.

Let us see what happens when we integrate carefully. Suppose separation leads to

$$\ln\lvert y\rvert = x + C.$$

Exponentiating: $\lvert y\rvert = e^{x+C} = e^C e^x$. Writing $A = e^C$ (so $A > 0$), we get $\lvert y\rvert = Ae^x$, that is,

$$y = \pm Ae^x.$$

Since A is an arbitrary positive constant, $\pm A$ is an arbitrary non-zero constant. Renaming it K, the general solution is $y = Ke^x$ with $K \in \mathbb{R} \setminus \{0\}$ (and $y = 0$ can usually be absorbed by allowing $K = 0$). The negative branch $y = -Ae^x$ would have been **lost** had we written $\ln y$ instead of $\ln\lvert y\rvert$.

> **WARNING** **Forgetting the Absolute Value**
>
> Writing $\ln y$ instead of $\ln\lvert y\rvert$ silently discards all negative solutions. In an examination, if the initial condition later gives $y < 0$, a candidate who dropped the absolute value will find no valid constant and lose the marks. Rule: every time you integrate $1/y$ (or $1/(ay + b)$), write the modulus signs immediately, and remove them only when exponentiating, absorbing the $\pm$ into the arbitrary constant.

3.4 Explicit versus Implicit General Solutions

After integration we obtain a relation $G(y) = F(x) + C$. Two situations arise.

Case 1: we can solve for y. If G is a simple expression (a power, a logarithm, an exponential), we isolate y and present the **general solution** explicitly as $y = \varphi(x, C)$. This is the preferred final form when the question says "express y in terms of x".

Case 2: we cannot (or need not) solve for y. Some integrations yield relations such as

$$\frac{y^2}{2} + \ln|y| = \frac{x^3}{3} + C,$$

which cannot be solved for y in closed form. The relation itself is the general solution, given in **implicit form**. This is perfectly acceptable — indeed, attempting to force an explicit form where none exists in elementary functions is a waste of examination time.

KEY POINTS **Tip: Read the Question's Final Instruction**

When the integral yields an implicit relation, solve for y only if the question requires it ("give y in terms of x", "show that $y = \ldots$"). Otherwise, a clean implicit relation with one arbitrary constant earns full marks. Never leave the answer as an unevaluated integral.

3.5 A Complete Worked Example, Step by Step

EXEMPLE **Solving $\frac{dy}{dx} = \frac{2x}{y}$**

Step 1 — Separate. Multiply both sides by y and by dx:

$$y\,dy = 2x\,dx.$$

Step 2 — Integrate.

$$\int y\,dy = \int 2x\,dx.$$

Step 3 — One constant.

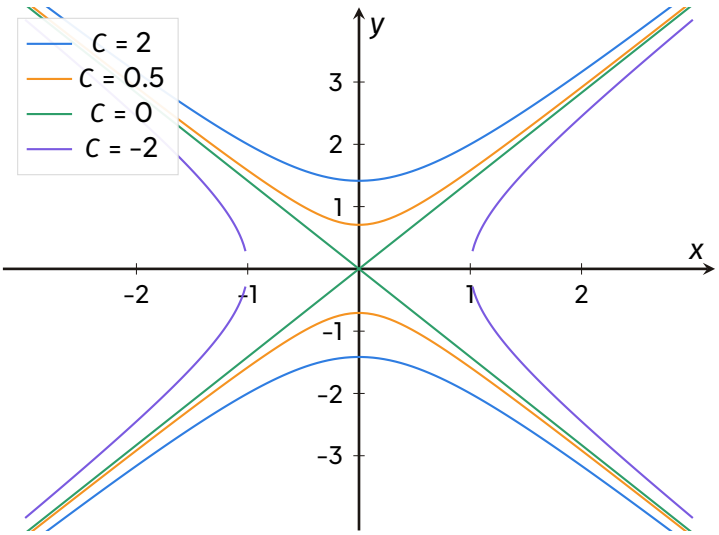
$$\frac{y^2}{2} = x^2 + C_1.$$

Step 4 — Simplify. Multiply by 2 and rename $2C_1$ as C:

$$y^2 = 2x^2 + C.$$

Step 5 — Explicit form (optional). $y = \pm\sqrt{2x^2 + C}$. The implicit form $y^2 = 2x^2 + C$ describes a family of hyperbolas (and the pair of straight lines $y = \pm\sqrt{2}\,x$ when $C = 0$); each value of C gives one curve of the family.

The figure below shows several members of this family. Notice the two branches (upper and lower) for each value of C: this is exactly the $\pm$ that the absolute-value discussion protects.



Family of solution curves $y^2 = 2x^2 + C$ of $\frac{dy}{dx} = \frac{2x}{y}$ for several values of C. Each value of the constant selects one curve; the general solution is the whole family. For $C = -2$ the hyperbola exists only where $2x^2 - 2 \geq 0$, i.e. $|x| \geq 1$.

3.6 Verification by Differentiation

A powerful habit — and one that costs thirty seconds — is to differentiate your answer and check that it satisfies the original equation. This is done by **implicit differentiation** when the solution is implicit.

For the example above, differentiate $y^2 = 2x^2 + C$ with respect to x:

$$2y\,\frac{dy}{dx} = 4x \quad\Longrightarrow\quad \frac{dy}{dx} = \frac{2x}{y},$$

which is exactly the given equation. The arbitrary constant C disappears on differentiation, confirming that the whole family satisfies the equation. **Always perform this check** before moving on in an examination: it catches sign errors, lost factors of 2, and forgotten constants instantly.

WORKED EXAMPLE 3 **Worked Exercise: an Equation Leading to a Logarithm**

Solve the differential equation $\frac{dy}{dx} = \frac{x}{x^2+1}\,y$, giving y explicitly in terms of x.

SOLUTION

Solution. The equation is separable with $f(x) = \frac{x}{x^2+1}$ and $g(y) = y$.

Step 1 — Separate (for $y \neq 0$):

$$\frac{dy}{y} = \frac{x}{x^2+1}\,dx.$$

Step 2 — Integrate. On the right, the numerator is (half of) the derivative of the denomi-

nator, so we use $\int \frac{f'(x)}{f(x)}\,dx = \ln|f(x)|$:

$$\int \frac{dy}{y} = \frac{1}{2}\int \frac{2x}{x^2+1}\,dx \quad\Longrightarrow\quad \ln|y| = \frac{1}{2}\ln(x^2+1) + C,$$

where the modulus is unnecessary on x^2+1 since it is always positive.
Step 3 — Exponentiate and handle the modulus.

$$|y| = e^C\,(x^2+1)^{1/2}.$$

Writing $K = \pm e^C$ (an arbitrary non-zero constant), and noting that $y = 0$ is recovered with $K = 0$:

$$\boxed{y = K\sqrt{x^2+1}, \qquad K \in \mathbb{R}.}$$

Verification. Differentiate: $\frac{dy}{dx} = K\cdot\frac{x}{\sqrt{x^2+1}} = \frac{x}{x^2+1}\cdot K\sqrt{x^2+1} = \frac{x}{x^2+1}\,y.$ ✓

WORKED EXAMPLE 4 **Worked Exercise: Trigonometric Separation, Implicit Answer**

Find the general solution of $\frac{dy}{dx} = \frac{\cos x}{\sin y}$, where $\sin y \neq 0$.

SOLUTION

Solution. Step 1 — Separate: $\sin y\,dy = \cos x\,dx$.
Step 2 — Integrate:

$$\int \sin y\,dy = \int \cos x\,dx \quad\Longrightarrow\quad -\cos y = \sin x + C_1.$$

Step 3 — Tidy: multiplying by -1 and renaming $-C_1$ as C:

$$\cos y + \sin x = C.$$

Here it is not natural to write y explicitly (one could write $y = \arccos(C - \sin x)$, but this introduces sign and branch complications). The implicit form $\cos y + \sin x = C$ is the expected general solution.
Verification. Differentiate implicitly: $-\sin y\,\frac{dy}{dx} + \cos x = 0$, so $\frac{dy}{dx} = \frac{\cos x}{\sin y}$. ✓

3.7 Common Mistakes and How to Avoid Them

WARNING **Traps in the Separation–Integration Process**

- Separating incorrectly. In $\frac{dy}{dx} = x + y$, you cannot separate — this equation is not separable (see Section 2). Never invent a separation that does not exist.
- Two constants. Writing $\frac{y^2}{2} + C_1 = x^2 + C_2$ and leaving both: merge them into one constant immediately.
- Dropping the modulus. $\int \frac{dy}{y} = \ln|y|$, always — until you exponentiate.
- Losing solutions when dividing. Dividing by $g(y)$ assumes $g(y) \neq 0$; check separately whether $g(y) = 0$ gives a (constant) solution, e.g. $y = 0$ above.
- Stopping at the integral. $\int \frac{dy}{y} = \int 2x\,dx$ is not an answer; evaluate both integrals.
- Forgetting to verify. One line of implicit differentiation protects all your marks.

EXERCISE 4 **Practice**

1. Find the general solution of $\frac{dy}{dx} = \frac{3x^2}{2y}$, expressing y explicitly in terms of x.
2. Find the general solution of $\frac{dy}{dx} = e^x y^2$, leaving your answer in the form $y = \varphi(x)$. What restriction must be placed on the constant?
3. Find the general solution of $\frac{dy}{dx} = \frac{1+y^2}{1+x^2}$ in implicit form. (Hint: $\int \frac{du}{1+u^2} = \arctan u$.)

SOLUTION **Exercise 1**

1. Separate: $2y\,dy = 3x^2\,dx$. Integrate: $y^2 = x^3 + C$. Hence $y = \pm\sqrt{x^3 + C}$.
2. Separate ($y \neq 0$): $\frac{dy}{y^2} = e^x\,dx$. Integrate: $-\frac{1}{y} = e^x + C$, so $y = -\frac{1}{e^x + C}$. We need $e^x + C \neq 0$ on the domain; also $y = 0$ is a separate constant solution lost when dividing by y^2.
3. Separate: $\frac{dy}{1+y^2} = \frac{dx}{1+x^2}$. Integrate: $\arctan y = \arctan x + C$. (Explicitly, $y = \tan(\arctan x + C)$, but the implicit form is the cleaner answer.)

KEY POINTS **Key Points of Section 3**

- Separation gives $\int \frac{dy}{g(y)} = \int f(x)\,dx + C$ — justified by integration by substitution.
- One constant of integration only, placed on one side.
- $\int \frac{dy}{y} = \ln|y|$: the modulus protects the negative branches; absorb ± into the constant when exponentiating.
- Give y explicitly if the question asks; otherwise a clean implicit relation is a complete answer.
- Always verify by differentiating your solution and recovering the original equation.

We can now produce the general solution of any separable equation: an entire family of curves indexed by C. But real problems — a cooling cup of tea in Bamenda, a rumour spreading through a school — pick out one curve from the family. Selecting that curve from a given starting value is the business of Section 4.

4 PARTICULAR SOLUTIONS USING INITIAL CONDITIONS

In the previous section we learnt how to separate the variables of a first order differential equation and integrate both sides to obtain the **general solution**. That general solution always contains an arbitrary constant C: it is not one curve but a whole family of curves. In real problems, however, we usually know one extra piece of information — for instance the population of a town at the moment we start counting, or the temperature of a pot of soup when it is taken off the fire. This extra information, called an **initial condition**, allows us to pick out exactly one curve from the family. That single curve is the **particular solution**. This section teaches you how to find it, how to choose between several candidate constants, and how to state the domain on which your answer is valid — all essential skills for Paper 2 of the GCE Advanced Level.

4.1 From a Family of Curves to a Single Curve: the Initial Condition

DEFINITION **Initial condition, initial value problem, particular solution**

Let $\frac{dy}{dx} = f(x)\,g(y)$ be a separable first order differential equation with general solution $F(x,y) = C$ (or $y = \varphi(x, C)$).

- An **initial condition** is a pair of values (x_0, y_0) such that the required solution satisfies $y(x_0) = y_0$. We often write "$y = y_0$ when $x = x_0$".
- A differential equation together with an initial condition is called an **initial value problem** (IVP).
- The **particular solution** of the IVP is the unique member of the family of general so-

lutions whose graph passes through the point (x_0, y_0). It is obtained by substituting $x = x_0, y = y_0$ into the general solution and solving for C.

Intuition.

Think of the general solution as a bus timetable listing all buses from Douala to Bamenda (infinitely many parallel curves). The initial condition tells you which bus you actually boarded: the one passing through your town at your time. Once C is fixed, there is no more freedom — the future and the past of the quantity are completely determined.

METHOD **Finding the particular solution**

1. Solve the equation by separation of variables and write the general solution, keeping the constant C visible.
2. Substitute the initial condition: replace x by x_0 and y by y_0.
3. Solve for C (or for the related constant $A = e^C$). If several values of the constant appear (for example ± from a square root), keep all candidates for the moment.
4. Select the branch compatible with the initial condition (sign, positivity, etc.).
5. State the final answer explicitly, in the form $y = \ldots$ (or as an implicit relation if y cannot be isolated), and mention the domain: an interval containing x_0 on which the solution is defined and satisfies the equation.

PROPERTY **Existence and uniqueness for separable equations**

Consider the initial value problem $\frac{dy}{dx} = f(x)\,g(y)$, $y(x_0) = y_0$. If f is continuous on an open interval containing x_0, and g is continuous with $g(y_0) \neq 0$ on an open interval containing y_0, then the IVP has exactly one solution, defined on some interval around x_0. Geometrically: through the point (x_0, y_0) passes one and only one solution curve, and two distinct solution curves never cross. (Statement only — the proof is not examinable at this level.)

WARNING **When $g(y_0) = 0$: the lost constant solutions**

If $g(y_0) = 0$, then $y = y_0$ is a constant solution of the equation (both sides of $\frac{dy}{dx} = f(x)g(y)$ vanish). Recall that constant solutions are often lost during separation because we divide by $g(y)$. Therefore: if the initial condition lies on a constant solution, that constant solution is the answer. For example, the IVP $\frac{dy}{dx} = ky$, $y(0) = 0$ has the solution $y = 0$ for all x — do not

try to force $0 = Ae^{kx}$ (which would require $A = 0$, giving the same answer, but for equations like $\frac{dy}{dx} = y^{2/3}$ the constant solution can be missed entirely). Always check the constant solutions before separating.

4.2 Worked Examples: Substituting the Initial Condition

WORKED EXAMPLE 5 **A first initial value problem**

Find the particular solution of $\frac{dy}{dx} = \frac{x}{y}$ satisfying $y(0) = 2$.

SOLUTION

Solution. Separating variables: $y\,\frac{dy}{dx} = x$, so

$$\int y\,dy = \int x\,dx \quad\Longrightarrow\quad \frac{y^2}{2} = \frac{x^2}{2} + C,$$

i.e. $y^2 = x^2 + 2C$. Writing $A = 2C$, the general solution is $y^2 = x^2 + A$.
Substitute $x = 0$, $y = 2$: $4 = 0 + A$, so $A = 4$.
Hence $y^2 = x^2 + 4$, and taking the square root gives $y = \pm\sqrt{x^2 + 4}$. Since $y(0) = 2 > 0$, we select the positive branch:

$$\boxed{y = \sqrt{x^2 + 4}}, \qquad x \in \mathbb{R}.$$

Check: $y(0) = \sqrt{4} = 2$ ✓ and $\frac{dy}{dx} = \frac{x}{\sqrt{x^2+4}} = \frac{x}{y}$ ✓.

WORKED EXAMPLE 6 **Population growth with given initial population**

A bacterial culture grows according to $\frac{dP}{dt} = 0.3P$, where P is the population (in thousands) and t the time in hours. Given that the initial population is $P(0) = 5$ (i.e. 5000 bacteria), find $P(t)$ and the population after 10 hours.

SOLUTION

Solution. Note $P = 0$ is a constant solution, but $P(0) = 5 \neq 0$, so we separate:

$$\int \frac{dP}{P} = \int 0.3\,dt \quad\Longrightarrow\quad \ln|P| = 0.3t + C.$$

Since $P > 0$, $P = e^{0.3t+C} = Ae^{0.3t}$ with $A = e^C > 0$.
Substitute $t = 0$, $P = 5$: $5 = Ae^0 = A$. Hence

$$\boxed{P(t) = 5e^{0.3t}}, \qquad t \geq 0.$$

After 10 hours: $P(10) = 5e^{3} \approx 5 \times 20.0855 \approx 100.4$ thousand, i.e. about 100 400 bacteria.

WORKED EXAMPLE 7 **Newton's law of cooling with initial temperature**

A pot of water is removed from the fire at temperature 95°C and placed in a kitchen at 25°C. By Newton's law of cooling, $\frac{dT}{dt} = -k(T - 25)$, where T is the temperature after t minutes. Given $k = 0.05$, find $T(t)$ and the temperature after 20 minutes.

SOLUTION

Solution. The constant solution $T = 25$ does not satisfy $T(0) = 95$, so we separate:

$$\int \frac{dT}{T - 25} = \int -0.05\,dt \quad \Longrightarrow \quad \ln|T - 25| = -0.05t + C.$$

Since $T > 25$, $T - 25 = Ae^{-0.05t}$, i.e. $T(t) = 25 + Ae^{-0.05t}$.
Substitute $t = 0$, $T = 95$: $95 = 25 + A$, so $A = 70$. Hence

$$\boxed{T(t) = 25 + 70e^{-0.05t}}, \qquad t \geq 0.$$

After 20 minutes: $T(20) = 25 + 70e^{-1} \approx 25 + 70(0.3679) \approx 50.8$°C.

4.3 Choosing the Correct Branch and Stating the Domain

When the integration produces $\ln|y|$, a square, or a square root, the constant may not be uniquely determined by algebra alone. The initial condition is the referee that decides between the candidates.

The ± ambiguity.

If the general solution is $y^2 = G(x) + A$, then $y = \pm\sqrt{G(x) + A}$. The sign is chosen so that $y(x_0) = y_0$: positive branch if $y_0 > 0$, negative branch if $y_0 < 0$. The two branches are different solutions and never meet (uniqueness theorem).

The sign of $A = e^{C}$.

From $\ln|y| = h(x) + C$ we get $|y| = e^{C}e^{h(x)}$, so $y = Ae^{h(x)}$ where A can be any nonzero real (positive or negative), and $A = 0$ recovers the constant solution $y = 0$. The initial condition fixes both the value and the sign of A.

Domain considerations.

A particular solution is only valid on an interval that

- **contains** x_0 (the solution must pass through the given point);
- avoids values where the formula breaks down (division by zero, logarithm of a non-positive number, square root of a negative number);
- is as large as possible subject to the above (the maximal interval of the solution).

WORKED EXAMPLE 8 **Branch selection and maximal domain**

Solve the IVP $\frac{dy}{dx} = \frac{1}{2y}$, $y(1) = -3$.

SOLUTION

Solution. Note $y = 0$ is excluded (division by $2y$), and $y(1) = -3 \neq 0$. Separating:

$$\int 2y\,dy = \int dx \quad \Longrightarrow \quad y^2 = x + C.$$

Substitute $x = 1$, $y = -3$: $9 = 1 + C$, so $C = 8$, giving $y^2 = x + 8$.
Since $y(1) = -3 < 0$, take the negative branch: $y = -\sqrt{x + 8}$.
Domain: we need $x + 8 > 0$ (strictly, since $y \neq 0$), i.e. $x > -8$, and this interval contains $x_0 = 1$.
Hence

$$\boxed{y = -\sqrt{x + 8}, \qquad x \in]-8, +\infty[}.$$

Check: $y(1) = -\sqrt{9} = -3$ ✓, $\frac{dy}{dx} = -\frac{1}{2\sqrt{x+8}} = \frac{1}{2y}$ ✓.

WORKED EXAMPLE 9 **Implicit solution and domain restriction**

Solve the IVP $\frac{dy}{dx} = \frac{2x}{y}$, $y(0) = -4$.

SOLUTION

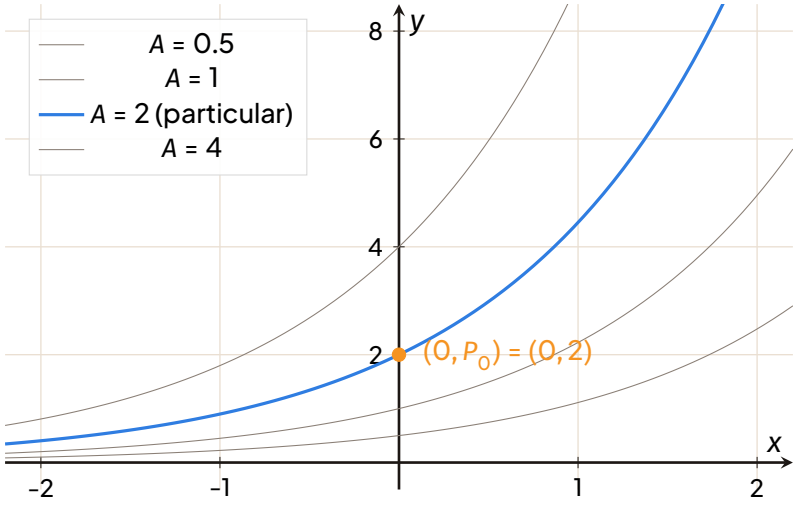
Solution. $y = 0$ is excluded; $y(0) = -4 \neq 0$. Separate: $y\,dy = 2x\,dx$, integrate: $\frac{y^2}{2} = x^2 + C$, so $y^2 = 2x^2 + A$. Substitute $x = 0$, $y = -4$: $16 = A$. Thus $y^2 = 2x^2 + 16$. Since $y(0) = -4 < 0$, we take $y = -\sqrt{2x^2 + 16}$. The expression under the square root is always positive, so the maximal domain is $\mathbb{R}$.

$$\boxed{y = -\sqrt{2x^2 + 16}, \qquad x \in \mathbb{R}}.$$

WARNING **Common mistakes with initial conditions**

- Forgetting to substitute before simplifying. Substitute (x_0, y_0) as soon as the general solution is found; do not carry C through extra manipulations.
- Taking the wrong branch. $y = \pm\sqrt{\ldots}$: the sign is dictated by the sign of y_0, not by convenience.
- Writing C instead of a number. The final answer must contain no arbitrary constant.
- Ignoring the domain. A formula like $y = -\sqrt{x + 8}$ stated for all x is wrong for $x < -8$.

- Missing the constant solution. If y_0 makes $g(y_0) = 0$, the answer is the constant function $y = y_0$; separation of variables is unnecessary.
- Forgetting the absolute value in $\ln|y|$. This leads to missing the negative branch when $y_0 < 0$.



Family of solutions $y = Ae^{0.8x}$ of $\frac{dy}{dx} = 0.8y$. The initial condition $y(0) = P_0 = 2$ selects the single curve $y = 2e^{0.8x}$ (thick blue) passing through $(0, 2)$.

DID YOU KNOW? **Why uniqueness matters**

The uniqueness theorem is the mathematical reason why prediction is possible. Because exactly one curve passes through (x_0, y_0), knowing the present state of a system (population, temperature, charge) and its law of change determines its entire future. This is the principle behind weather forecasting models, the design of cooling systems, and the modelling of epidemic spread studied by Cameroonian health researchers.

KEY POINTS **Key points**

- An initial condition $y(x_0) = y_0$ fixes the constant C: substitute, solve, rewrite.
- If algebra gives several candidates ($\pm$, sign of A), the initial condition selects the correct branch.
- If $g(y_0) = 0$, the particular solution is the constant solution $y = y_0$.

- Always finish with an explicit answer $y = \ldots$ (or a clean implicit relation) and the domain — an interval containing x_0.
- Always check: does $y(x_0) = y_0$? Does the function satisfy the differential equation?

4.4 Practice: Exam-Style Initial Value Problems

EXERCISE 5 **Exercise 1 — direct substitution**

Find the particular solution of $\frac{dy}{dx} = 3x^2e^{-y}$ given that $y = 0$ when $x = 0$.

SOLUTION **Exercise 1**

Separate: $\int e^y\,dy = \int 3x^2\,dx$, so $e^y = x^3 + C$. Substitute $x = 0$, $y = 0$: $e^0 = 1 = 0 + C$, so $C = 1$. Hence $e^y = x^3 + 1$, i.e. $y = \ln(x^3 + 1)$, valid where $x^3 + 1 > 0$, i.e. $x > -1$. Since $x_0 = 0 \in]-1, +\infty[$, the solution is $y = \ln(x^3 + 1)$, $x > -1$.

EXERCISE 6 **Exercise 2 — exponential decay**

The mass m (in grams) of a radioactive substance satisfies $\frac{dm}{dt} = -0.02m$, with t in days. Initially the mass is 80 g. Find $m(t)$ and the mass remaining after 30 days.

SOLUTION **Exercise 2**

$m = 0$ is a constant solution but $m(0) = 80 \neq 0$. Separate: $\ln|m| = -0.02t + C$, so $m = Ae^{-0.02t}$. Substitute $t = 0$, $m = 80$: $A = 80$. Hence $m(t) = 80e^{-0.02t}$, $t \geq 0$. After 30 days: $m(30) = 80e^{-0.6} \approx 80(0.5488) \approx 43.9$ g.

EXERCISE 7 **Exercise 3 — branch selection**

Solve the IVP $\frac{dy}{dx} = \frac{x}{y^2}$, $y(0) = -1$.

SOLUTION **Exercise 3**

$y = 0$ is excluded by the equation itself; $y(0) = -1 \neq 0$. Separate: $\int y^2\,dy = \int x\,dx$, so $\frac{y^3}{3} = \frac{x^2}{2} + C$, i.e. $y^3 = \frac{3}{2}x^2 + A$. Substitute $x = 0$, $y = -1$: $(-1)^3 = A$, so $A = -1$. Hence $y^3 = \frac{3}{2}x^2 - 1$ and $y = \sqrt[3]{\frac{3}{2}x^2 - 1}$ (the cube root needs no sign choice and automatically gives $y(0) = -1$). Domain: all real x, since the cube root is defined everywhere; the solution is $y = \sqrt[3]{\frac{3x^2}{2} - 1}$, $x \in \mathbb{R}$.

EXERCISE 8 **Exercise 4 — trigonometric initial condition**

Find the particular solution of $\frac{dy}{dx} = y\cos x$ with $y\left(\frac{\pi}{2}\right) = e$.

SOLUTION **Exercise 4**

$y = 0$ is a constant solution but $y(\pi/2) = e \neq 0$. Separate: $\int \frac{dy}{y} = \int \cos x\,dx$, so $\ln|y| = \sin x + C$, i.e. $y = Ae^{\sin x}$. Substitute $x = \frac{\pi}{2}$, $y = e$: $e = Ae^{\sin(\pi/2)} = Ae^1$, so $A = 1$. Hence $y = e^{\sin x}$, $x \in \mathbb{R}$. Check: $y(\pi/2) = e^1 = e$ ✓.

EXERCISE 9 **Exercise 5 — cooling problem with two conditions**

A body cools according to Newton's law $\frac{dT}{dt} = -k(T - 20)$, where the room temperature is 20°C. The body's initial temperature is $T(0) = 100$°C, and after 10 minutes it has fallen to 60°C. Determine k, write $T(t)$, and find when the temperature reaches 30°C.

SOLUTION **Exercise 5**

Separate: $\ln|T - 20| = -kt + C$, so $T = 20 + Ae^{-kt}$. First condition $T(0) = 100$: $100 = 20 + A$, so $A = 80$, giving $T = 20 + 80e^{-kt}$. Second condition $T(10) = 60$: $60 = 20 + 80e^{-10k}$, so $e^{-10k} = \frac{40}{80} = \frac{1}{2}$, hence $-10k = \ln\frac{1}{2} = -\ln 2$ and $k = \frac{\ln 2}{10} \approx 0.0693$. Thus $T(t) = 20 + 80e^{-\frac{\ln 2}{10}t} = 20 + 80\left(\frac{1}{2}\right)^{t/10}$, $t \geq 0$. For $T = 30$: $30 = 20 + 80e^{-kt}$, so $e^{-kt} = \frac{10}{80} = \frac{1}{8} = \left(\frac{1}{2}\right)^3$, giving $\frac{t}{10} = 3$, i.e. $t = 30$ minutes.

EXERCISE 10 **Exercise 6 — rational function with restricted domain**

Solve the IVP $\frac{dy}{dx} = \frac{y^2}{x}$, $y(1) = 2$.

SOLUTION **Exercise 6**

Note $y = 0$ is a constant solution, but $y(1) = 2 \neq 0$. Separate: $\int \frac{dy}{y^2} = \int \frac{dx}{x}$, so $-\frac{1}{y} = \ln|x| + C$. Substitute $x = 1$, $y = 2$: $-\frac{1}{2} = \ln 1 + C = C$. Hence $-\frac{1}{y} = \ln|x| - \frac{1}{2}$. Solve for y: $\frac{1}{y} = \frac{1}{2} - \ln|x|$, so $y = \frac{1}{\frac{1}{2} - \ln|x|} = \frac{2}{1 - 2\ln|x|}$. Domain: we need $x \neq 0$ (from $\ln|x|$) and $1 - 2\ln|x| \neq 0 \Rightarrow \ln|x| \neq \frac{1}{2} \Rightarrow |x| \neq \sqrt{e}$. The initial point is $x_0 = 1 > 0$, so we take the interval $(0, \sqrt{e})$ containing 1.

$$\boxed{y = \frac{2}{1 - 2\ln x}, \qquad x \in]0, \sqrt{e}[}.$$

5 APPLICATIONS OF SEPARABLE DIFFERENTIAL EQUATIONS AND EXAM PREPARATION

In the previous section we learnt how to pin down the constant of integration using an initial condition, so that a whole family of curves is reduced to the single curve passing through a given point. We now put this skill to work where it truly matters: real situations. Almost every differential equation you will meet at the GCE Advanced Level comes wrapped in a story — a population of bacteria, a cooling cup of tea, a radioactive sample, a tank of brine. The examination does not only test whether you can integrate; it tests whether you can translate a story into mathematics, solve it, and interpret the answer. This section trains exactly that skill, model by model, and ends with examination strategy.

5.1 Exponential Growth and Decay: $\frac{dP}{dt} = kP$

Intuition. Many quantities grow (or shrink) at a rate proportional to their current size: a bacterial colony doubles faster when it is larger, because more cells are dividing; a radioactive sample decays faster when more unstable nuclei remain; money in a savings account earns interest on the interest already earned. In each case the rate of change is proportional to the quantity itself. This single observation produces the most important differential equation of the whole syllabus.

PROPERTY **Exponential growth/decay model**

If a quantity $P(t)$ satisfies

$$\frac{dP}{dt} = kP,$$

where k is a constant, then its general solution is

$$P(t) = P_0\, e^{kt}, \qquad P_0 = P(0).$$

If $k > 0$ we have **exponential growth**; if $k < 0$ we have **exponential decay**. Writing $k = -\lambda$ with $\lambda > 0$ for decay, the **half-life** (time for the quantity to be halved) is

$$t_{1/2} = \frac{\ln 2}{\lambda}.$$

For growth ($k > 0$), the **doubling time** is $t_2 = \dfrac{\ln 2}{k}$.

Derivation (examinable). Separating variables:

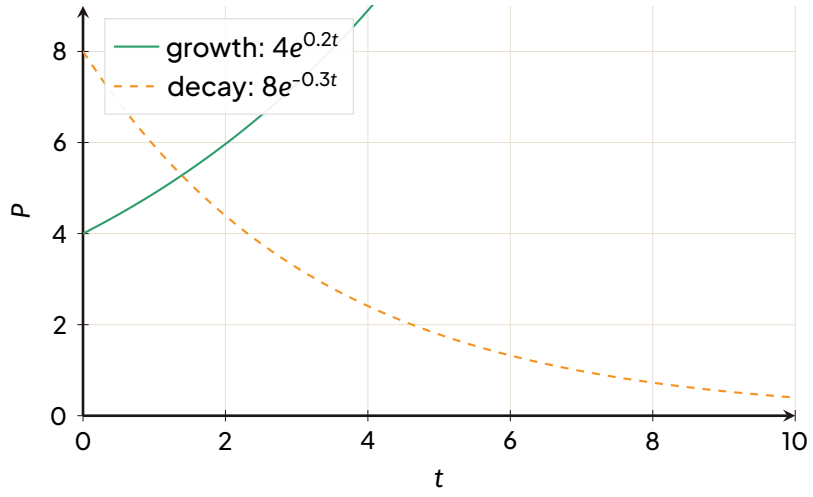
$$\frac{dP}{P} = k\,dt \implies \ln|P| = kt + C \implies P = Ae^{kt}, \quad A = \pm e^{C}.$$

The initial condition $P(0) = P_0$ gives $A = P_0$.

Derivation of the half-life. For decay, $P = P_0 e^{-\lambda t}$. The half-life is defined by $P(t_{1/2}) = \dfrac{P_0}{2}$:

$$\frac{P_0}{2} = P_0 e^{-\lambda t_{1/2}} \implies e^{-\lambda t_{1/2}} = \frac{1}{2} \implies -\lambda t_{1/2} = -\ln 2 \implies t_{1/2} = \frac{\ln 2}{\lambda}.$$

Note that P_0 cancels: the half-life does not depend on the initial amount. This is the signature of exponential decay, and it is why half-life is such a useful concept.



Exponential growth ($k > 0$) curves away from the axis ever faster; exponential decay ($k < 0$) approaches zero asymptotically, losing the same fraction in every equal time interval.

WORKED EXAMPLE 10 **Radioactive decay**

A sample contains $N_0 = 8.0 \times 10^6$ radioactive nuclei and decays according to $\frac{dN}{dt} = -\lambda N$. After 10 days, only half remain. Find λ and the number of nuclei after 30 days.

SOLUTION

Model. The solution is $N = N_0 e^{-\lambda t}$.
Find λ. The half-life condition gives

$$\frac{N_0}{2} = N_0 e^{-10\lambda} \implies \lambda = \frac{\ln 2}{10} \approx 0.0693\ \text{day}^{-1}.$$

Evaluate. After 30 days:

$$N(30) = 8.0 \times 10^6\, e^{-30\lambda} = 8.0 \times 10^6\, e^{-3\ln 2} = 8.0 \times 10^6 \times \frac{1}{8} = 1.0 \times 10^6.$$

Check. 30 days is three half-lives, so the amount is divided by $2^3 = 8$: the answer is consistent.

WORKED EXAMPLE 11 **Bacterial growth**

A culture of bacteria grows according to $\frac{dP}{dt} = kP$. Initially there are 2000 bacteria; after 3 hours there are 16000. Find k and the size of the culture after 5 hours.

SOLUTION

Model. $P(t) = 2000e^{kt}$.
Find k. At $t = 3$: $16000 = 2000e^{3k}$, so $e^{3k} = 8$, giving

$$k = \frac{\ln 8}{3} = \frac{3\ln 2}{3} = \ln 2 \approx 0.693\ \text{h}^{-1}.$$

So the population doubles every hour.
Evaluate. $P(5) = 2000e^{5\ln 2} = 2000 \times 2^5 = 64000$ bacteria.
Check. After 5 hours there have been 5 doublings from 2000: $2000 \to 4000 \to 8000 \to 16000 \to 32000 \to 64000$. Consistent.

5.2 Newton's Law of Cooling

Intuition. A hot bowl of achu left on the table does not cool at a constant rate: it cools quickly at first (when it is much hotter than the room) and ever more slowly as its temperature approaches room temperature. Newton observed that the rate of cooling is proportional to the difference between the object's temperature and the ambient temperature.

PROPERTY **Newton's law of cooling**

If $T(t)$ is the temperature of a body placed in an environment at constant temperature T_{env}, then

$$\frac{dT}{dt} = -k\left(T - T_{\text{env}}\right), \qquad k > 0,$$

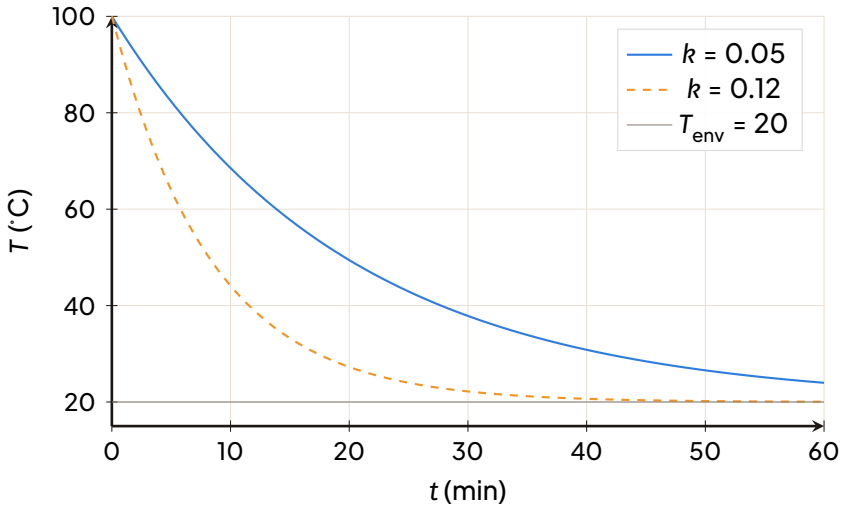
with solution

$$T(t) = T_{\text{env}} + \left(T_0 - T_{\text{env}}\right)e^{-kt}.$$

Derivation. Let $u = T - T_{\text{env}}$. Since T_{env} is constant, $\frac{du}{dt} = \frac{dT}{dt} = -ku$, which is exponential decay of u. Hence $u = u_0 e^{-kt}$, i.e. $T - T_{\text{env}} = (T_0 - T_{\text{env}})e^{-kt}$.

WARNING **Cool down towards, not to, zero**

The body never cools below room temperature, and its temperature tends to T_{env}, not to 0. The equation must therefore contain $T - T_{\text{env}}$, not T alone. Writing $\frac{dT}{dt} = -kT$ would predict that your tea eventually freezes in a warm room — a clear sign that the model is wrong.



Cooling curves $T(t) = T_{\text{env}} + (T_0 - T_{\text{env}})e^{-kt}$ with $T_0 = 100$, $T_{\text{env}} = 20$. Larger k means faster cooling; both curves approach T_{env} asymptotically.

WORKED EXAMPLE 12 **Cooling of a liquid**

A liquid at 90°C is placed in a room at 25°C. After 10 minutes its temperature is 60°C. Find

its temperature after 25 minutes.

SOLUTION

Model. $T(t) = 25 + 65e^{-kt}$ since $T_0 - T_{\text{env}} = 90 - 25 = 65$.
Find k. At $t = 10$: $60 = 25 + 65e^{-10k}$, so $e^{-10k} = \dfrac{35}{65} = \dfrac{7}{13}$, giving

$$k = \frac{1}{10}\ln\frac{13}{7} \approx 0.0619\ \text{min}^{-1}.$$

Evaluate. At $t = 25$:

$$T(25) = 25 + 65e^{-25k} = 25 + 65\left(\frac{7}{13}\right)^{2.5} \approx 25 + 65(0.2127) \approx 38.8^{\circ}\text{C}.$$

Check. The temperature lies between 60°C and 25°C and is still decreasing — consistent with the physics.

5.3 Chemical Reaction Rates

In a reaction $A \to$ products, the **rate law** often takes the form

$$\frac{dC}{dt} = -kC^n,$$

where $C(t)$ is the concentration of reactant A and n is the **order** of the reaction.

First-order kinetics ($n = 1$). $\dfrac{dC}{dt} = -kC$ gives $C = C_0e^{-kt}$: exponential decay, with half-life $\dfrac{\ln 2}{k}$ independent of C_0.

Second-order kinetics ($n = 2$). $\dfrac{dC}{dt} = -kC^2$. Separating:

$$-\frac{dC}{C^2} = k\,dt \implies \frac{1}{C} = kt + C' \implies \frac{1}{C} = kt + \frac{1}{C_0},$$

using $C(0) = C_0$. So a plot of $\dfrac{1}{C}$ against t is a straight line of slope k — a standard way of testing second-order behaviour experimentally.

EXEMPLE **Second-order reaction**

A reactant obeys $\dfrac{dC}{dt} = -0.02\,C^2$ (concentration in mol L^{-1}, time in minutes) with $C_0 = 0.5$ mol L^{-1}. Then

$$\frac{1}{C} = 0.02t + \frac{1}{0.5} = 0.02t + 2 \implies C(t) = \frac{1}{2 + 0.02t}.$$

At $t = 50$ min: $C = \dfrac{1}{3} \approx 0.333$ mol L^{-1}.

WARNING **Units of the rate constant**

The units of k depend on the order: for $n = 1$, k is in time^{-1}; for $n = 2$, k is in concentration^{-1} time^{-1}. Examiners award marks for correct units.

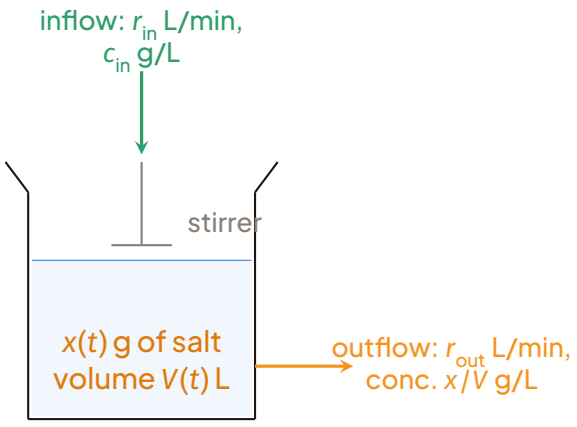
5.4 Mixing Problems

The set-up. A tank contains a solution (e.g. brine). A solution flows in at a given rate with a given concentration, the mixture is kept uniform by stirring, and flows out at a given rate. We want the mass $x(t)$ of solute (salt) in the tank.

The governing principle is a simple balance:

$$\frac{dx}{dt} = \text{(rate of solute in)} - \text{(rate of solute out)}.$$

The outflow concentration is $\dfrac{x(t)}{V(t)}$, where $V(t)$ is the volume of liquid in the tank at time t, because the stirring keeps the mixture uniform.



Mixing tank: solute enters at rate $r_{in}c_{in}$ and leaves at rate $r_{out}\,x/V(t)$.

PROPERTY **Mixing equation**

With the notation above,

$$\frac{dx}{dt} = r_{in}c_{in} - r_{out}\frac{x}{V(t)}, \qquad V(t) = V_0 + (r_{in} - r_{out})t.$$

When $r_{in} = r_{out}$ (constant volume) and c_{in} is constant, this equation is **separable**: it has the form $\dfrac{dx}{dt} = a - bx$. When the volume varies, the equation generally requires an integrating factor — beyond this chapter — but the balance equation itself must still be set up correctly.

WARNING **Volume may change**

If $r_{in} \neq r_{out}$, the volume is not constant. Always write $V(t) = V_0 + (r_{in} - r_{out})t$ before attempting to separate variables. Using V_0 throughout when the volume changes is the most common error in mixing questions.

WORKED EXAMPLE 13 **Brine tank, constant volume**

A tank contains 100 L of brine with 20 g of dissolved salt. Brine containing 0.5 g/L flows in at 4 L/min, and the well-stirred mixture flows out at 4 L/min. Find the mass of salt after 25 minutes.

SOLUTION

Set up. Volume constant: $V = 100$ L. Rate in: $4 \times 0.5 = 2$ g/min. Rate out: $4 \times \frac{x}{100} = \frac{x}{25}$ g/min. Hence

$$\frac{dx}{dt} = 2 - \frac{x}{25} = \frac{50 - x}{25}, \qquad x(0) = 20.$$

Separate and integrate.

$$\frac{dx}{50 - x} = \frac{dt}{25} \implies -\ln|50 - x| = \frac{t}{25} + C \implies 50 - x = Ae^{-t/25}.$$

Initial condition. $x(0) = 20 \Rightarrow A = 30$, so $x(t) = 50 - 30e^{-t/25}$.
Evaluate. $x(25) = 50 - 30e^{-1} \approx 50 - 11.04 \approx 38.96$ g.
Interpret. As $t \to \infty$, $x \to 50$ g, i.e. the concentration tends to $\frac{50}{100} = 0.5$ g/L, the inflow concentration — exactly as expected physically.

METHOD **Translating a word problem into a separable DE**

1. Identify the unknown quantity and its variable (e.g. $P(t)$, $T(t)$, $x(t)$), with units.
2. Identify the rate statement in the text: "proportional to", "rate of change", "half of" etc., and name the proportionality constant k.
3. Write the balance or rate equation $\frac{dy}{dt} = \ldots$ and check it is separable: the right-hand side must be a product (or quotient) of a function of y and a function of t.
4. Separate, integrate, and apply the initial condition to find the constant.
5. Answer the question asked (a value, a time, a limit) and interpret it in the context, with units.

5.5 Exam Strategies and Common Mistakes

KEY POINTS **GCE exam strategy for DE questions**

- Read twice. Underline the rate statement and the initial condition; note all units.
- Identify the model: growth/decay ($y' = ky$), cooling ($T' = -k(T - T_{\text{env}})$), reaction ($C' = -kC^n$), mixing (balance equation).
- Show the separation step explicitly. Marks are awarded for correct separation even if the subsequent integration contains a minor slip.
- Use the initial condition immediately after integrating to fix the constant.
- Interpret and check: does the sign make sense? Does the long-term limit agree with the physics? Are the units right?

WARNING **Frequent traps**

- Forgetting the constant of integration, or applying the initial condition before integrating.
- Sign errors in decay problems: write $\frac{dP}{dt} = -kP$ with $k > 0$ stated explicitly, rather than leaving the sign of k ambiguous.
- In cooling problems, writing $\frac{dT}{dt} = -kT$ instead of $-k(T - T_{\text{env}})$.
- In mixing problems, using the inflow concentration instead of x/V for the outflow.
- Giving a final answer without units or without rounding sensibly.

EXERCISE 11 **Practice**

1. A population grows according to $\frac{dP}{dt} = 0.03P$ with $P(0) = 5000$. Find the time needed for the population to double.
2. A body cools from 80°C to 50°C in 15 minutes in a room at 20°C. When will it reach 30°C?
3. A tank of 200 L contains 10 g of salt. Pure water enters at 5 L/min and the mixture leaves at 5 L/min. Find $x(t)$ and the mass of salt after 40 minutes.

SOLUTION **Exercise 1**

$P = 5000e^{0.03t}$. Doubling: $e^{0.03t} = 2 \Rightarrow t = \dfrac{\ln 2}{0.03} \approx 23.1$ time units.

SOLUTION **Exercise 2**

$T = 20 + 60e^{-kt}$; $50 = 20 + 60e^{-15k} \Rightarrow e^{-15k} = \frac{1}{2}$, so $k = \frac{\ln 2}{15}$. Then $30 = 20 + 60e^{-kt} \Rightarrow e^{-kt} = \frac{1}{6} \Rightarrow t = \dfrac{\ln 6}{k} = \dfrac{15 \ln 6}{\ln 2} \approx 38.8$ min.

SOLUTION **Exercise 3**

Rate in $= 0$; rate out $= \dfrac{5x}{200} = \dfrac{x}{40}$. So $\dfrac{dx}{dt} = -\dfrac{x}{40}$, giving $x = 10e^{-t/40}$. At $t = 40$: $x = 10e^{-1} \approx 3.68$ g.

DID YOU KNOW? **Why this chapter matters beyond the exam**

Separable equations are the workhorses of quantitative science: pharmacists use them for drug elimination from the bloodstream, engineers for capacitor discharge, demographers for population projections in Cameroon's development planning, and nuclear physicists for carbon dating. Mastering the translation from words to equations is a skill you will reuse throughout higher education.

6 INTEGRATION ACTIVITY

6.1 Aims of the activity

This activity brings together all the skills developed in this chapter in a single, authentic modelling task. Working in groups of three or four, you will:

- **Recognise** a first order differential equation with separable variables and justify why the method of separation applies;
- **Separate the variables** correctly, taking care with a right-hand side that depends on time through given data;
- **Integrate** both sides to obtain a **general solution**, handling a piecewise-constant coefficient by splitting the integral over sub-intervals;
- **Use an initial condition** to determine the constant of integration and hence a **particular solution**;
- **Interpret** the solution: predict a harvest date, test the sensitivity of the prediction to a parameter, and discuss the limitations of the model.

By the end of the activity you should be able to move fluently through the complete modelling cycle: equation → separation → integration → initial condition → interpretation and critique. This cycle is exactly what the GCE Advanced Level examiner rewards in structured modelling questions.

6.2 Situation

Designing a Sustainable Harvest Plan for a Cocoa Farm.

The Nkongsamba area, in the Littoral Region of Cameroon, is one of the country's important cocoa-producing zones. A cooperative of smallholder farmers wants to plan the main harvest so that picking starts when the pods are ripe but before over-ripening and pod disease reduce the quality of the beans. The cooperative's agricultural adviser proposes the following simplified model for the **ripeness index** $R(t)$ of an average pod, where t is measured in months from the start of the growing season:

$$\frac{dR}{dt} = k\big(T(t) - T_{\text{opt}}\big)\,R,$$

where:

- $R(t)$ is the ripeness index, a number between 0 (green pod) and 1 (fully ripe);
- $T(t)$ is the average monthly temperature, in °C;
- $T_{\text{opt}} = 22$ °C is a reference temperature below which ripening effectively stops;
- $k = 0.06$ per month per °C is a growth constant estimated from previous seasons.

The adviser explains that the model says: the rate of ripening is proportional to the current ripeness and to the excess of temperature above the reference value.

Records from the cooperative's weather station give the following average monthly temperatures for the six-month growing season:

Month n	1	2	3	4	5	6
Period	$0 \le t < 1$	$1 \le t < 2$	$2 \le t < 3$	$3 \le t < 4$	$4 \le t < 5$	$5 \le t \le 6$
T (°C)	24	25	26	26	25	24

At the start of the season ($t = 0$), field measurements give $R(0) = 0.1$. The cooperative considers pods **harvest-ready** when $R = 0.9$.

Your group's mission: produce a ripeness formula $R(t)$, predict the harvest month, and advise the cooperative on how reliable the prediction is.

6.3 Tasks

Work through the following questions in your group. Keep a written record: your group will present its solution to the class.

1. **Recognition.** Write the model in the form $\frac{dR}{dt} = f(t)\,g(R)$, identifying $f(t)$ and $g(R)$ explicitly. Hence explain why the equation is a first order differential equation with **separable variables**. What is the order of the equation, and why?

2. **Separation.** Assuming $R > 0$, separate the variables and write down the equation you obtain after integrating both sides with respect to their own variable. State clearly where the constant of integration appears.

3. **Integration with piecewise data.** Because $T(t)$ is constant on each month, show that for t in month n,
$$\ln R = k\sum_{j=1}^{n-1}(T_j - T_{\text{opt}}) + k(T_n - T_{\text{opt}})(t - (n-1)) + C,$$
where T_j is the temperature in month j. (Hint: split $\int_0^t k(T(s) - T_{\text{opt}})\,ds$ month by month.)

4. **Initial condition.** Use $R(0) = 0.1$ to find C, and hence write the particular solution in the form
$$R(t) = 0.1\exp\left(k\int_0^t \left(T(s) - T_{\text{opt}}\right)ds\right).$$

5. **Computing the accumulated heat.** Complete the following table of the accumulated exponent $A(t) = k\int_0^t (T(s) - T_{\text{opt}})\,ds$ at the end of each month:

t (months)	0	1	2	3	4	5	6
$A(t)$							
$R(t)$							

6. **Harvest prediction.** Determine the value of t for which $R(t) = 0.9$. In which month should the cooperative begin the harvest? Give the answer as a number of months (to 2 decimal places) and interpret it in words.

7. **Sensitivity.** The constant $k = 0.06$ is only an estimate. Repeat the harvest-time calculation with $k = 0.05$ and $k = 0.07$. Comment on how sensitive the harvest date is to k, and what this means for the cooperative's planning.
8. **Critical reflection.** State at least **three** assumptions or weaknesses of the model (think about what happens when $T < T_{\text{opt}}$, about rainfall, pests, variation between pods, and the meaning of $R > 1$). Suggest one improvement.
9. **Presentation.** Prepare a five-minute group presentation: one member justifies separability, one presents the integration, one the harvest prediction and sensitivity, and one the critique.

6.4 Model answer

1. Recognition.

The equation $\dfrac{dR}{dt} = k\big(T(t) - T_{\text{opt}}\big)R$ has the form $\dfrac{dR}{dt} = f(t)\,g(R)$ with

$$f(t) = k\big(T(t) - T_{\text{opt}}\big), \qquad g(R) = R.$$

The highest derivative present is the first derivative, so the equation is of **first order**. Since the right-hand side is a product of a function of t alone and a function of R alone, the variables are **separable**.

2. Separation.

Since $R > 0$ we may divide both sides by R:

$$\frac{1}{R}\frac{dR}{dt} = k\big(T(t) - T_{\text{opt}}\big) \quad \Longrightarrow \quad \int \frac{dR}{R} = \int k\big(T(t) - T_{\text{opt}}\big)\,dt.$$

Integrating the left side gives $\ln R$ (as $R > 0$, no absolute value is needed); the constant C is placed on the right:

$$\ln R = \int_0^t k\big(T(s) - T_{\text{opt}}\big)\,ds + C.$$

3. Integration with piecewise data.

On month j, $T(s) = T_j$ is constant, so over that month the integrand contributes $k(T_j - T_{\text{opt}})$ per unit time. For t in month n, split the integral into the completed months $1, \dots, n-1$ (each of length 1) plus the partial current month of length $t - (n-1)$:

$$\int_0^t k(T(s) - T_{\text{opt}})\,ds = k\sum_{j=1}^{n-1}(T_j - T_{\text{opt}})\cdot 1 + k(T_n - T_{\text{opt}})\big(t - (n-1)\big),$$

which gives the stated formula.

4. Initial condition.

Setting $t = 0$: the integral from 0 to 0 is 0, so $\ln R(0) = C$, giving $C = \ln 0.1$. Therefore

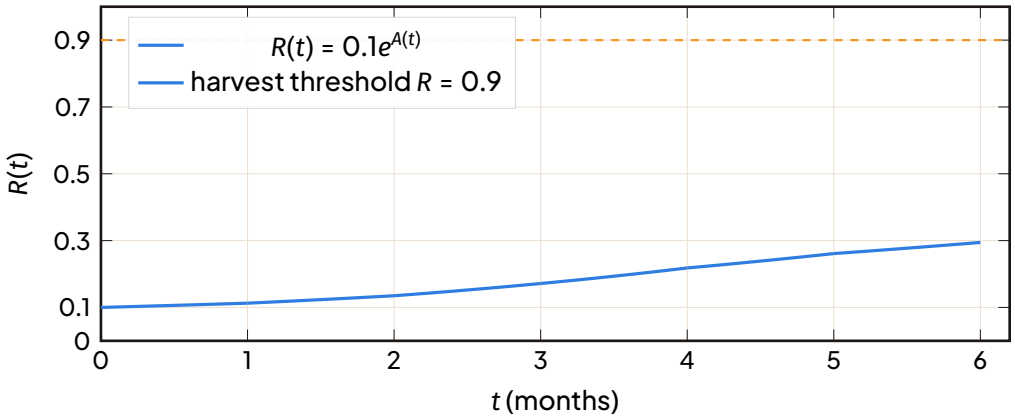
$$\ln R = \ln 0.1 + k\int_0^t \big(T(s) - T_{\text{opt}}\big)\,ds \quad \Longrightarrow \quad R(t) = 0.1\exp\left(k\int_0^t \big(T(s) - T_{\text{opt}}\big)\,ds\right).$$

5. Accumulated heat.

With $k = 0.06$ and $T_{opt} = 22$: the monthly excesses $T_j - 22$ are 2, 3, 4, 4, 3, 2. Multiplying by k gives monthly increments 0.12, 0.18, 0.24, 0.24, 0.18, 0.12. Accumulating and computing $R = 0.1e^{A(t)}$:

t (months)	0	1	2	3	4	5	6
$A(t)$	0	0.12	0.30	0.54	0.78	0.96	1.08
$R(t)$	0.100	0.113	0.135	0.172	0.218	0.261	0.294

For example, $R(3) = 0.1e^{0.54} \approx 0.1 \times 1.716 = 0.172$.



The ripeness index grows exponentially within each month, but stays far below the harvest threshold $R = 0.9$ throughout the six-month season.

6. Harvest prediction.

We need $R(t) = 0.9$, i.e. $0.1e^{A(t)} = 0.9$, so

$$A(t) = \ln 9 \approx 2.1972.$$

But the table shows $A(6) = 1.08 < 2.1972$: within the six-month season the accumulated exponent is far too small, and R only reaches about 0.29. **Conclusion: under this model with $k = 0.06$, the pods do not reach harvest ripeness within six months.** Solving formally, with the seasonal pattern repeating, the average monthly increment of A is $\frac{1.08}{6} = 0.18$, so $A(t) \approx 0.18\,t$ and

$$t \approx \frac{\ln 9}{0.18} \approx 12.21 \text{ months}.$$

The model therefore predicts harvest-readiness after roughly **12.2 months**, i.e. only in the following growing season — a signal that the model (or the value of k) needs revision, since real cocoa pods mature within a single season.

7. Sensitivity.

Repeating with the average monthly increment scaled by k:

- $k = 0.05$: increment $= \frac{0.05}{0.06} \times 0.18 = 0.15$, so $t \approx \frac{2.1972}{0.15} \approx 14.65$ months;
- $k = 0.07$: increment $= \frac{0.07}{0.06} \times 0.18 = 0.21$, so $t \approx \frac{2.1972}{0.21} \approx 10.46$ months.

A change of about 17% in k shifts the predicted harvest time by roughly two months in either direction: the prediction is **quite sensitive** to k. The cooperative should therefore treat k as the key quantity to measure more accurately, e.g. by recording ripeness over one full season and fitting the model.

8. Critical reflection.

Weaknesses of the model include:

- If $T < T_{\text{opt}}$, the model gives $\frac{dR}{dt} < 0$: pods would "un-ripen", which is biologically unrealistic — ripening should at worst stop;
- rainfall, humidity, sunlight and soil fertility are ignored, though they strongly affect cocoa development;
- all pods are described by a single index $R(t)$, ignoring variation between trees and pods;
- nothing prevents R from exceeding 1, although the index is defined only up to 1;
- k and T_{opt} are rough estimates, and monthly average temperatures hide daily fluctuations.

One improvement: replace the factor $(T - T_{\text{opt}})$ by $\max(T - T_{\text{opt}}, 0)$ so ripening never reverses, and use weekly temperature data for a finer integration.

9. Presentation.

Groups present their work; the class compares harvest predictions and discusses which assumption is most damaging to the model's credibility.

WARNING **Common traps in this activity**

Do not integrate $k(T(t) - T_{\text{opt}})$ as if T were a single constant — you must split the integral month by month. Do not forget that $\ln R$ requires $R > 0$ (justified here by $R(0) = 0.1 > 0$). Finally, when the model contradicts reality (harvest after 12 months), say so honestly: recognising a model's failure is a valued modelling skill, not a mistake.

7 METHODS AND WORKED EXAMPLES

7.1 Skill 1: Verifying that a Function is a Solution of a Differential Equation

A **differential equation** is an equation that involves an unknown function y and one or more of its derivatives. When only the first derivative $\frac{dy}{dx}$ appears, the equation is said to be of **first order**. A **solution** of such an equation is a function $y = f(x)$ which, together with its derivative, satisfies the equation identically on some interval. Before learning how to find solutions, we must be able to check a proposed solution — a skill that the GCE tests directly and that rewards careful, methodical work.

METHOD **Verifying a Proposed Solution**

To check whether a given function $y = f(x)$ is a solution of the differential equation $F(x, y, y') = 0$:

1. Compute the derivative $y' = f'(x)$ (and y'' if needed).
2. Substitute y and y' into the left-hand side of the equation.
3. Simplify carefully and compare with the right-hand side.
4. If the equality holds for all x in the domain, conclude that f is a solution; otherwise it is not.
5. If an initial condition $y(x_0) = y_0$ is given, check it separately by direct substitution.

Reflex: never "solve" anything here — you only verify. This is a frequent source of easy marks at the GCE.

WORKED EXAMPLE 14 **Verifying a Solution**

Show that the function $y = 3e^{2x}$ is a solution of the differential equation $\frac{dy}{dx} - 2y = 0$, and that it satisfies the initial condition $y(0) = 3$.

SOLUTION

Solution.
Step 1: differentiate. Given $y = 3e^{2x}$, we obtain

$$\frac{dy}{dx} = 3 \times 2e^{2x} = 6e^{2x}.$$

Step 2: substitute into the left-hand side.

$$\frac{dy}{dx} - 2y = 6e^{2x} - 2\left(3e^{2x}\right) = 6e^{2x} - 6e^{2x} = 0.$$

Step 3: conclude. The left-hand side equals the right-hand side for every real x, so $y = 3e^{2x}$ is a solution of the equation on $\mathbb{R}$.
Step 4: check the initial condition. $y(0) = 3e^{0} = 3 \times 1 = 3$. The condition $y(0) = 3$ is satisfied.
Hence $y = 3e^{2x}$ is the particular solution of $\dfrac{dy}{dx} - 2y = 0$ satisfying $y(0) = 3$.

WORKED EXAMPLE 15 **Verifying an Implicitly Defined Solution**

Show that the relation $x^2 + y^2 = 25$ defines a solution of the differential equation $y\dfrac{dy}{dx} + x = 0$.

SOLUTION

Solution.
Step 1: differentiate implicitly with respect to x:

$$\frac{d}{dx}(x^2 + y^2) = \frac{d}{dx}(25) \quad \Longrightarrow \quad 2x + 2y\frac{dy}{dx} = 0.$$

Step 2: simplify. Dividing by 2:

$$y\frac{dy}{dx} + x = 0,$$

which is exactly the given equation. Hence every function y defined implicitly by $x^2 + y^2 = 25$ (i.e. $y = \pm\sqrt{25 - x^2}$ on $-5 < x < 5$) is a solution. Geometrically, the solution curves are semicircles centred at the origin.

7.2 Skill 2: Recognising a Separable Equation

Not every first-order equation can be solved by the techniques of this chapter. The crucial first step is to recognise those equations whose right-hand side splits into a part depending only on x and a part depending only on y.

DEFINITION **Separable First-Order Equation**

A first-order differential equation is called **separable** (or "with separable variables") if it can be written in the form

$$\frac{dy}{dx} = g(x)\,h(y),$$

where g is a function of x alone and h is a function of y alone.

METHOD **Testing for Separability**

Practical reflexes:

1. Try to factorise the right-hand side: look for common factors, e.g. $xy + x = x(y + 1)$.
2. Powers multiply across variables: $e^{x+y} = e^x e^y$ is separable; $e^x + y$ is not.
3. Sums that cannot be factorised, e.g. $\frac{dy}{dx} = x + y$, are not separable.
4. Equations of the form $\frac{dy}{dx} = g(x)$ (no y) and $\frac{dy}{dx} = h(y)$ (no x) are trivially separable.

WORKED EXAMPLE 16 **Recognising Separable Equations**

For each equation, state whether it is separable. If it is, write it in the form $\frac{dy}{dx} = g(x)\,h(y)$.

1. $\frac{dy}{dx} = xy^2 + x$ (b) $\frac{dy}{dx} = e^{x-y}$ (c) $\frac{dy}{dx} = x^2 + y^2$ (d) $\frac{dy}{dx} = \frac{\sin x}{y}$

SOLUTION

Solution.
(a) Factorise the right-hand side: $xy^2 + x = x(y^2 + 1)$. This is a product of $g(x) = x$ and $h(y) = y^2 + 1$. Separable.
(b) Using the law of indices, $e^{x-y} = e^x\, e^{-y} = e^x \cdot \frac{1}{e^y}$. With $g(x) = e^x$ and $h(y) = e^{-y}$, the equation is separable.
(c) $x^2 + y^2$ cannot be written as a product of a function of x alone and a function of y alone (any attempted factorisation introduces cross terms in xy). Not separable.
(d) $\frac{\sin x}{y} = \sin x \cdot \frac{1}{y}$, a product of $g(x) = \sin x$ and $h(y) = \frac{1}{y}$. Separable.

WARNING **Sums versus Products**

The single most common error is to treat $\frac{dy}{dx} = x + y$ as separable. A sum of an x-term and a y-term is not separable unless it factorises. Always ask: "Can I write the right-hand side as one factor in x times one factor in y?"

7.3 Skill 3: Separating Variables and Finding a General Solution (Polynomial/Algebraic Cases)

METHOD **Solving a Separable Equation — General Solution**

1. Write the equation as $\frac{dy}{dx} = g(x)\,h(y)$.
2. Separate: divide by $h(y)$ (noting any constant solutions where $h(y) = 0$) and multiply by dx:
$$\frac{dy}{h(y)} = g(x)\,dx.$$
3. Integrate both sides: $\int \frac{dy}{h(y)} = \int g(x)\,dx$.
4. Add one constant of integration only (a single C covers both sides).
5. If possible, make y the subject to obtain the **general solution** explicitly; otherwise leave it implicit.

Key formulas: $\int x^n\,dx = \frac{x^{n+1}}{n+1}\,(n \neq -1)$, $\int \frac{dy}{y} = \ln|y|$, $\int e^{ax}\,dx = \frac{1}{a}e^{ax}$.

KEY POINTS **Why One Constant is Enough**

Integrating both sides formally gives $F(y) + C_1 = G(x) + C_2$. Subtracting, $F(y) = G(x) + (C_2 - C_1)$, and since $C_2 - C_1$ is just an arbitrary constant, we rename it C. Writing two constants is not wrong, but failing to merge them loses presentation marks.

WORKED EXAMPLE 17 **General Solution of a Separable Equation**

Find the general solution of the differential equation

$$\frac{dy}{dx} = \frac{x}{y}, \qquad y \neq 0.$$

SOLUTION

Solution.
Step 1: identify. The right-hand side is $x \cdot \frac{1}{y}$, a product of a function of x and a function of y: the equation is separable.
Step 2: separate the variables. Multiply both sides by $y\,dx$:

$$y\,dy = x\,dx.$$

Step 3: integrate both sides.

$$\int y\,dy = \int x\,dx \quad\Longrightarrow\quad \frac{y^2}{2} = \frac{x^2}{2} + C_1.$$

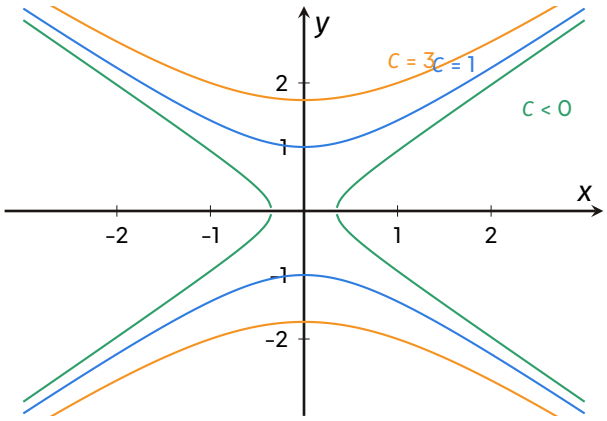
Step 4: simplify. Multiplying through by 2 and writing $C = 2C_1$:

$$y^2 = x^2 + C.$$

Step 5: make y the subject.

$$y = \pm\sqrt{x^2 + C}.$$

Conclusion. The general solution is $y^2 = x^2 + C$, i.e. $y = \pm\sqrt{x^2 + C}$, where C is an arbitrary constant. Geometrically, the solution curves are rectangular hyperbolas $y^2 - x^2 = C$.
Check: differentiating $y^2 = x^2 + C$ implicitly gives $2y\frac{dy}{dx} = 2x$, i.e. $\frac{dy}{dx} = \frac{x}{y}$. ✓



The family of solution curves $y^2 = x^2 + C$ of $\frac{dy}{dx} = \frac{x}{y}$: one arbitrary constant, one family of curves.

7.4 Skill 4: Equations Leading to Logarithms and Exponentials

METHOD **Handling $\int \frac{dy}{y}$ and Exponential Answers**

When separation produces $\int \frac{dy}{y}$:

1. Integrate to $\ln|y| = \int g(x)\,dx + C$.
2. Exponentiate: $|y| = e^{\int g(x)dx + C} = e^C\, e^{\int g(x)dx}$.
3. Absorb the sign and the constant: write $y = A\,e^{\int g(x)dx}$ where $A = \pm e^C$ is an arbitrary non-

zero constant; the case $A = 0$ (the zero solution) is usually added by inspection.

4. For $\frac{dy}{dx} = ky$, memorise the model result: $y = Ae^{kx}$.

WARNING **Forgetting the Constant or the Absolute Value**

Do not write $\ln y$ instead of $\ln|y|$, and do not add a constant on both sides — one constant is enough. At the GCE, a missing constant costs marks even if everything else is correct.

PROPERTY **The Exponential Model (proof examinable)**

The general solution of $\frac{dy}{dx} = ky$ (where k is a constant) is

$$y = Ae^{kx}, \qquad A \in \mathbb{R}.$$

Proof. For $y \neq 0$, separation gives $\int \frac{dy}{y} = \int k\,dx$, so $\ln|y| = kx + C$. Exponentiating, $|y| = e^C e^{kx}$, hence $y = \pm e^C e^{kx}$. Setting $A = \pm e^C$ and adjoining the zero solution $y = 0$ (which corresponds to $A = 0$) gives $y = Ae^{kx}$, $A \in \mathbb{R}$. ■

WORKED EXAMPLE 18 **Exponential-Type General Solution**

Find the general solution of

$$\frac{dy}{dx} = \frac{3x^2}{y}, \qquad y > 0,$$

and the general solution of $\frac{dy}{dx} = 2xy$.

SOLUTION

Solution.
First equation. Separate: $y\,dy = 3x^2\,dx$. Integrate:

$$\int y\,dy = \int 3x^2\,dx \quad \Longrightarrow \quad \frac{y^2}{2} = x^3 + C_1.$$

Hence $y^2 = 2x^3 + C$ (with $C = 2C_1$), and since $y > 0$:

$$y = \sqrt{2x^3 + C}.$$

Second equation. $\frac{dy}{dx} = 2xy = 2x \cdot y$ is separable. Note $y = 0$ is a constant solution. For

$y \neq 0$, divide by y and multiply by dx:

$$\frac{dy}{y} = 2x\,dx.$$

Integrate:

$$\int \frac{dy}{y} = \int 2x\,dx \quad \Longrightarrow \quad \ln|y| = x^2 + C.$$

Exponentiate:

$$|y| = e^{x^2+C} = e^C e^{x^2} \quad \Longrightarrow \quad y = \pm e^C e^{x^2}.$$

Writing $A = \pm e^C$ (and allowing $A = 0$ to include the zero solution):

$$y = Ae^{x^2}, \quad A \in \mathbb{R}.$$

Check: $\frac{dy}{dx} = A \cdot 2xe^{x^2} = 2x\left(Ae^{x^2}\right) = 2xy$. ✓

7.5 Skill 5: Trigonometric Separable Equations

METHOD **Separable Equations with Trigonometric Functions**

1. Separate as usual; typical integrals are

$$\int \sin x\,dx = -\cos x, \qquad \int \sec^2 x\,dx = \tan x, \qquad \int \tan x\,dx = -\ln|\cos x| = \ln|\sec x|.$$

2. GCE questions usually arrange for $\int \cot y\,dy = \ln|\sin y|$ or $\int \tan y\,dy = -\ln|\cos y|$.

3. Remember $\int \frac{f'(y)}{f(y)}\,dy = \ln|f(y)|$ — the "derivative over function" pattern is the key reflex.

WORKED EXAMPLE 19 **A Trigonometric Equation**

Find the general solution of

$$\frac{dy}{dx} = e^x \cos^2 y, \qquad \cos y \neq 0.$$

SOLUTION

Solution.
Step 1: separability. $e^x \cos^2 y = e^x \cdot \cos^2 y$: product of a function of x and a function of y.

Separable.
Step 2: separate. Divide by $\cos^2 y$ (valid since $\cos y \neq 0$) and multiply by dx:

$$\frac{dy}{\cos^2 y} = e^x\,dx \quad \Longleftrightarrow \quad \sec^2 y\,dy = e^x\,dx.$$

Step 3: integrate. Since $\int \sec^2 y\,dy = \tan y$ and $\int e^x\,dx = e^x$:

$$\tan y = e^x + C.$$

Step 4: make y the subject.

$$y = \tan^{-1}\left(e^x + C\right).$$

Conclusion. The general solution is $y = \arctan\left(e^x + C\right)$, $C \in \mathbb{R}$.
Check: differentiate: $\frac{dy}{dx} = \frac{e^x}{1 + (e^x + C)^2}$. Also $\cos^2 y = \frac{1}{1 + \tan^2 y} = \frac{1}{1 + (e^x + C)^2}$, so $e^x \cos^2 y = \frac{e^x}{1 + (e^x + C)^2} = \frac{dy}{dx}$. ✓

WORKED EXAMPLE 20 **An Equation Leading to** $\ln|\sin y|$

Find the general solution of $\frac{dy}{dx} = \frac{\cos y}{\sin y}\,x^2$, where $\sin y \neq 0$.

SOLUTION

Solution.
Step 1: separate. The right-hand side is $x^2 \cdot \cot y$: separable. Divide by $\cot y$:

$$\tan y\,dy = x^2\,dx.$$

Step 2: integrate. Using $\int \tan y\,dy = -\ln|\cos y|$:

$$-\ln|\cos y| = \frac{x^3}{3} + C_1.$$

Step 3: simplify. Multiply by -1 and write $C = -C_1$:

$$\ln|\cos y| = C - \frac{x^3}{3}.$$

Exponentiating, $|\cos y| = e^C e^{-x^3/3}$, so the general solution can be written

$$\cos y = Ae^{-x^3/3}, \qquad A \neq 0,$$

or equivalently $y = \arccos\left(Ae^{-x^3/3}\right)$ where defined.
Check: differentiating $\cos y = Ae^{-x^3/3}$ implicitly gives $-\sin y\,\frac{dy}{dx} = -x^2 Ae^{-x^3/3} = -x^2 \cos y$, hence $\frac{dy}{dx} = x^2\frac{\cos y}{\sin y}$. ✓

7.6 Skill 6: Finding Particular Solutions from Initial Conditions

The general solution contains an arbitrary constant and therefore describes a whole family of curves. An **initial condition** $y(x_0) = y_0$ picks out the single member of the family passing through the point (x_0, y_0); the result is called a **particular solution**.

METHOD **From General to Particular Solution**

1. Solve the equation to obtain the general solution (with constant C or A).
2. Substitute $x = x_0$ and $y = y_0$ into the general solution.
3. Solve the resulting equation for the constant.
4. Rewrite the solution with this value: this is the **particular solution**.
5. State the domain of validity if relevant (e.g. where a square root or logarithm is defined).

Reflex: if the general solution is $y = Ae^{f(x)}$ and $y(x_0) = y_0$, then $A = y_0 e^{-f(x_0)}$.

WORKED EXAMPLE 21 **Particular Solution (GCE Style)**

1. Find the general solution of $\dfrac{dy}{dx} = \dfrac{y}{x}$, for $x > 0, y > 0$.
2. Hence find the particular solution for which $y = 6$ when $x = 2$.

SOLUTION

Solution.
(a) General solution. The equation is separable: $\dfrac{y}{x} = \dfrac{1}{x} \cdot y$. Separate:

$$\frac{dy}{y} = \frac{dx}{x}.$$

Integrate both sides:

$$\ln y = \ln x + C \qquad \text{(valid since } x > 0,\ y > 0\text{)}.$$

Exponentiate:

$$y = e^{\ln x + C} = e^C\, x.$$

Writing $A = e^C > 0$:

$$y = Ax.$$

(The solution curves are straight lines through the origin.)
(b) Particular solution. Substitute $x = 2, y = 6$ into $y = Ax$:

$$6 = 2A \quad \Longrightarrow \quad A = 3.$$

Conclusion. The particular solution is

$$y = 3x, \qquad x > 0.$$

Check: $\frac{dy}{dx} = 3$ and $\frac{y}{x} = \frac{3x}{x} = 3$. ✓ Also $y(2) = 6$. ✓

WORKED EXAMPLE 22 **Particular Solution with a Square Root**

Find the particular solution of $\frac{dy}{dx} = \frac{x}{y}$ $(y > 0)$ for which $y = 5$ when $x = 0$, and state the largest interval on which it is defined.

SOLUTION

Solution.
Step 1: general solution. From Skill 3, $y^2 = x^2 + C$, and since $y > 0$, $y = \sqrt{x^2 + C}$.
Step 2: apply the condition. $y(0) = 5$ gives $\sqrt{C} = 5$, so $C = 25$.
Step 3: particular solution.

$$y = \sqrt{x^2 + 25}.$$

Step 4: domain. Since $x^2 + 25 > 0$ for every real x, the solution is defined on the whole of $\mathbb{R}$.
Check: $\frac{dy}{dx} = \frac{x}{\sqrt{x^2 + 25}} = \frac{x}{y}$. ✓

7.7 Skill 7: Solving Application Problems (Modelling with Separable Equations)

Many natural and economic quantities change at a rate that depends on the quantity itself: populations, radioactive masses, temperatures, the value of an investment. Such situations are modelled by separable equations, which is why this chapter matters far beyond the examination hall.

METHOD **Modelling with Separable Differential Equations**

1. Translate the situation into an equation: "rate of change proportional to the quantity present" gives $\frac{dQ}{dt} = kQ$; "proportional to the difference from a limiting value L" gives $\frac{dQ}{dt} = k(L - Q)$.
2. Identify the variables and the initial condition $Q(0) = Q_0$.
3. Separate, integrate, find the general solution.

4. Use the initial condition to fix the constant; use any extra data point to fix k.
5. Answer the question asked (a time, a quantity, a percentage), giving units and sensible rounding.

Key models: exponential growth/decay $Q = Q_0 e^{kt}$; Newton's law of cooling $T - T_{\text{env}} = (T_0 - T_{\text{env}})e^{-kt}$.

WORKED EXAMPLE 23 **Radioactive Decay and Newton Cooling**

(a) A radioactive substance decays so that the mass m (in grams) satisfies $\frac{dm}{dt} = -km$. Initially the mass is 80 g, and after 5 days it is 60 g. Find the mass remaining after 10 days.
(b) A bottle at 90 °C is placed in a room at 30 °C. Its temperature T satisfies $\frac{dT}{dt} = -k(T - 30)$. After 10 minutes the temperature is 60 °C. Find k and the temperature after 20 minutes.

SOLUTION

Solution.
(a) General solution. Separate: $\frac{dm}{m} = -k\,dt$. Integrate: $\ln m = -kt + C$, so $m = Ae^{-kt}$.
Initial condition. $m(0) = 80 \Rightarrow A = 80$, hence $m = 80e^{-kt}$.
Find k. $m(5) = 60$: $60 = 80e^{-5k} \Rightarrow e^{-5k} = \frac{3}{4} \Rightarrow -5k = \ln\frac{3}{4}$, so $k = \frac{1}{5}\ln\frac{4}{3} \approx 0.0575$ per day.
Mass after 10 days. Note $e^{-10k} = \left(e^{-5k}\right)^2 = \left(\frac{3}{4}\right)^2 = \frac{9}{16}$:

$$m(10) = 80 \times \frac{9}{16} = 45 \text{ g}.$$

(b) General solution. Let $u = T - 30$; then $\frac{du}{dt} = -ku$, so $u = Ae^{-kt}$, i.e.

$$T = 30 + Ae^{-kt}.$$

Initial condition. $T(0) = 90 \Rightarrow A = 60$, hence $T = 30 + 60e^{-kt}$.
Find k. $T(10) = 60$: $60 = 30 + 60e^{-10k} \Rightarrow e^{-10k} = \frac{30}{60} = \frac{1}{2}$, so

$$k = \frac{\ln 2}{10} \approx 0.0693 \text{ min}^{-1}.$$

Temperature after 20 minutes. $e^{-20k} = \left(e^{-10k}\right)^2 = \frac{1}{4}$:

$$T(20) = 30 + 60 \times \frac{1}{4} = 30 + 15 = 45\,^{\circ}\text{C}.$$

Interpretation: the temperature gap halves every 10 minutes — the hallmark of exponential decay.

DID YOU KNOW? **Why "Half-Life" is Constant**

In part (a) above, the mass fell from 80 g to 60 g in 5 days, and from 60 g to 45 g in the next 5 days: the ratio $\frac{3}{4}$ is the same over each equal time interval. This is the defining feature of exponential decay $m = m_0 e^{-kt}$: the time needed to halve the quantity (the **half-life**) is always $T_{1/2} = \dfrac{\ln 2}{k}$, independent of the starting amount. This principle underlies radiocarbon dating and the management of radioactive waste.

7.8 Skill 8: Full GCE-Style Structured Question

METHOD **Tackling a Structured Examination Question**

1. Read the whole question first; part (a) usually feeds into part (b).
2. Show every line of the separation and integration — method marks are awarded for $\int \frac{dy}{h(y)} = \int g(x)\,dx$ even if a later slip occurs.
3. Keep the constant symbolic until the initial condition is applied.
4. In the final part, substitute numbers after obtaining the exact formula, and give the final answer to the accuracy requested.

WORKED EXAMPLE 24 **Comprehensive GCE-Style Exercise**

The number of bacteria N in a culture grows according to

$$\frac{dN}{dt} = \frac{N}{2}\left(1 + \frac{1}{t+1}\right), \qquad t \geq 0,$$

where t is measured in hours. Initially $N = 1000$.

1. Show that $\dfrac{1}{N}\,dN = \dfrac{t+2}{2(t+1)}\,dt$.
2. Hence show that $\ln N = \dfrac{t}{2} + \dfrac{1}{2}\ln(t+1) + C$ for some constant C.
3. Find the particular solution for N in terms of t.
4. Calculate, to the nearest whole number, the value of N when $t = 3$.

SOLUTION

Solution.

(a) The equation is separable. Divide by N and multiply by dt:

$$\frac{dN}{N} = \frac{1}{2}\left(1 + \frac{1}{t+1}\right)dt = \frac{1}{2}\cdot\frac{(t+1)+1}{t+1}\,dt = \frac{t+2}{2(t+1)}\,dt. \quad \blacksquare$$

(b) Write $\dfrac{t+2}{t+1} = \dfrac{(t+1)+1}{t+1} = 1 + \dfrac{1}{t+1}$. Integrate both sides:

$$\int \frac{dN}{N} = \frac{1}{2}\int\left(1 + \frac{1}{t+1}\right)dt.$$

$$\ln N = \frac{1}{2}\big(t + \ln(t+1)\big) + C = \frac{t}{2} + \frac{1}{2}\ln(t+1) + C. \quad \blacksquare$$

(Here $N > 0$ and $t + 1 > 0$, so no absolute-value signs are needed.)
(c) Apply $N = 1000$ when $t = 0$:

$$\ln 1000 = 0 + \tfrac{1}{2}\ln 1 + C \quad\Longrightarrow\quad C = \ln 1000.$$

Therefore

$$\ln N = \frac{t}{2} + \frac{1}{2}\ln(t+1) + \ln 1000.$$

Exponentiating and using $e^{\frac{1}{2}\ln(t+1)} = \sqrt{t+1}$:

$$N = 1000\,e^{t/2}\sqrt{t+1}.$$

(d) At $t = 3$:

$$N = 1000\,e^{1.5}\sqrt{4} = 2000\,e^{1.5} \approx 2000 \times 4.4817 \approx 8963.$$

Conclusion: after 3 hours there are approximately **8963** bacteria.

WARNING **Frequent Errors at the GCE**

- Dividing by $h(y)$ without checking whether $h(y) = 0$ gives extra (constant) solutions.
- Writing $\ln y$ when y could be negative: use $\ln|y|$.
- Adding two constants C_1 and C_2 and forgetting to merge them.
- Applying the initial condition before integrating — the condition is used after the general solution is found.
- In applications, mixing up the proportionality constant sign: decay requires $-k$ with $k > 0$.

EXERCISE 12 Practice — Core Skills

1. Verify that $y = 2e^{-3x}$ satisfies $\frac{dy}{dx} + 3y = 0$ with $y(0) = 2$.
2. State which of the following are separable, giving $g(x)$ and $h(y)$ where possible:
   (a) $\frac{dy}{dx} = x^2y$ (b) $\frac{dy}{dx} = x + xy$ (c) $\frac{dy}{dx} = \ln(x + y)$ (d) $\frac{dy}{dx} = \frac{y}{x^2 + 1}$.
3. Find the general solution of $\frac{dy}{dx} = \frac{x^2}{y^2}$, $y \neq 0$.
4. Find the general solution of $\frac{dy}{dx} = 3x^2y$, and the particular solution with $y(0) = 4$.
5. A quantity P grows according to $\frac{dP}{dt} = 0.2\,P$ with $P(0) = 500$. Find P when $t = 5$, giving your answer to the nearest whole number.

SOLUTION Practice — Core Skills

1. $y' = -6e^{-3x}$, so $y' + 3y = -6e^{-3x} + 6e^{-3x} = 0$ for all x; and $y(0) = 2e^0 = 2$. Both conditions hold. ✓
2. (a) Separable: $g(x) = x^2$, $h(y) = y$. (b) Separable: $x + xy = x(1+y)$, so $g(x) = x$, $h(y) = 1+y$. (c) Not separable (a sum inside the logarithm cannot be split into a product). (d) Separable: $g(x) = \frac{1}{x^2 + 1}$, $h(y) = y$.
3. Separate: $y^2\,dy = x^2\,dx$. Integrate: $\frac{y^3}{3} = \frac{x^3}{3} + C_1$, so $y^3 = x^3 + C$ and $y = \sqrt[3]{x^3 + C}$.
4. Separate: $\frac{dy}{y} = 3x^2\,dx$ (noting $y = 0$ is a constant solution). Integrate: $\ln|y| = x^3 + C$, so $y = Ae^{x^3}$. Condition $y(0) = 4$: $A = 4$. Particular solution: $y = 4e^{x^3}$.
5. Model solution: $P = 500e^{0.2t}$. At $t = 5$: $P = 500e^1 \approx 500 \times 2.71828 \approx 1359$.

EXERCISE 13 GCE-Style Consolidation

A tank contains water leaking out so that the volume V (in litres) satisfies

$$\frac{dV}{dt} = -\frac{V}{t + 4}, \qquad t \geq 0,$$

where t is in minutes. Initially $V = 200$ litres.

1. Show that $\ln V = C - \ln(t + 4)$ for some constant C.
2. Hence show that $V = \frac{800}{t + 4}$.

3. Find the volume after 6 minutes, and the time at which the volume falls to 40 litres.

SOLUTION **GCE-Style Consolidation**

(a) Separate: $\frac{dV}{V} = -\frac{dt}{t+4}$. Integrate: $\ln V = -\ln(t+4) + C$ (valid since $V > 0, t+4 > 0$). ■

(b) Exponentiate: $V = e^{C}(t+4)^{-1} = \frac{A}{t+4}$ with $A = e^{C}$. Condition $V(0) = 200$: $200 = \frac{A}{4}$, so $A = 800$ and $V = \frac{800}{t+4}$. ■

(c) At $t = 6$: $V = \frac{800}{10} = 80$ litres. For $V = 40$: $40 = \frac{800}{t+4} \Rightarrow t+4 = 20 \Rightarrow t = 16$ minutes.

8 EXERCISES

8.1 I check my knowledge

EXERCISE 14 **Multiple choice: identifying a separable equation**

Only one of the following differential equations is separable. Which one? Justify your answer by writing the right-hand side as a product (or quotient) of a function of x alone and a function of y alone.

1. $\frac{dy}{dx} = x + y$
2. $\frac{dy}{dx} = e^{x+y}$
3. $\frac{dy}{dx} = \sin(x + y)$
4. $\frac{dy}{dx} = xy + x + y + 1$

EXERCISE 15 **True or false?**

State whether each assertion is true or false, justifying your answer carefully.

1. Every first-order differential equation of the form $\frac{dy}{dx} = f(x)\,g(y)$ is separable.
2. The equation $\frac{dy}{dx} = 3y$ has general solution $y = Ce^{3x}$, where C is an arbitrary constant.
3. The function $y = 2e^{x^2}$ is a solution of the initial-value problem $\frac{dy}{dx} = 2xy$, $y(0) = 2$.
4. When separating variables in $\frac{dy}{dx} = y^2$, dividing by y^2 may cause us to lose the constant solution $y = 0$.

EXERCISE 16 **Definitions**

1. Define a **first-order differential equation** and give one example.
2. Explain what is meant by the **general solution** and by a **particular solution** of a first-order differential equation.

3. What is an **initial condition**, and why does a first-order equation need exactly one initial condition to determine a unique particular solution?

EXERCISE 17 **Direct calculations**

Find the general solution of each of the following separable equations.

1. $\dfrac{dy}{dx} = \dfrac{x}{y}$
2. $\dfrac{dy}{dx} = e^{2x}\,y$
3. $\dfrac{dy}{dx} = \dfrac{1}{x^2 y}, \quad x > 0, y > 0$
4. $x\dfrac{dy}{dx} = y + 1, \quad y > -1$

8.2 I practise

EXERCISE 18 **An initial-value problem with an explicit solution**

Solve the initial-value problem

$$\frac{dy}{dx} = \frac{3x^2}{2y}, \qquad y(1) = 2.$$

Give the explicit solution (i.e. y expressed in terms of x), state its domain of validity, and explain how the initial condition determines the sign of the square root.

EXERCISE 19 **Exponential growth of a bacterial culture**

A bacterial culture in a laboratory in Yaoundé grows according to the law

$$\frac{dN}{dt} = 0.05\,N,$$

where N is the number of bacteria (in thousands) and t is the time in hours. Given that $N(0) = 500$:

1. Find $N(t)$.
2. Calculate, to the nearest hour, the time required for the population to triple.
3. Show that this tripling time does not depend on the initial population.

EXERCISE 20 **Trigonometric and logarithmic separation**

1. Find the general solution of $\frac{dy}{dx} = y\cos x$ and sketch, on the same axes, the solution curves passing through $(0, 1)$, $(0, -1)$ and $(0, 0)$.
2. Solve the initial-value problem $\frac{dy}{dx} = \frac{y \ln x}{x}$, $y(1) = 3$, for $x > 0$.

EXERCISE 21 **Newton's law of cooling**

A cup of tea at 90°C is placed in a room kept at 20°C. According to Newton's law of cooling,

$$\frac{dT}{dt} = -k(T - 20),$$

where T is the temperature of the tea in degrees Celsius, t the time in minutes and $k > 0$ a constant. After 5 minutes the tea is at 70°C.

1. Solve the differential equation and determine the value of k.
2. Find the temperature of the tea after 15 minutes, giving your answer to the nearest degree.
3. What happens to $T(t)$ as $t \to +\infty$? Comment.

8.3 I prepare for the GCE

EXERCISE 22 **GCE-style structured question: curve through a point**

(10 marks) A curve passes through the point $(1, 2)$ and is such that the gradient at every point (x, y) of the curve is equal to $\frac{2y}{x}$.

(a) Write down a differential equation satisfied by y and show that it is separable. (2 marks)

(b) Solve the differential equation, giving the general solution in the form $y = f(x)$. (4 marks)

(c) Use the given point to find the particular solution, and state the set of values of x for which it is valid. (2 marks)

(d) Find the equation of the tangent to the curve at the point where $x = 2$. (2 marks)

EXERCISE 23 **GCE-style structured question: a mixing problem**

(12 marks) A tank at a water-treatment station in Douala initially contains 100 litres of pure water. Brine containing 0.2 kg of salt per litre flows into the tank at a rate of 3 litres per minute, and the well-stirred mixture flows out at a rate of 2 litres per minute. Let $S(t)$ be the amount of salt, in kilograms, in the tank after t minutes.

(a) Show that the volume of liquid in the tank after t minutes is $V(t) = 100 + t$ litres. (1 mark)

(b) Explain why

$$\frac{dS}{dt} = 0.6 - \frac{2S}{100 + t},$$

and show that this equation is not separable as it stands. (3 marks)

(c) Now suppose instead that the mixture flows out at 3 litres per minute (so the volume stays constant at 100 litres). Write down the new differential equation for $S(t)$, show that it is separable, and solve it given that $S(0) = 0$. (5 marks)

(d) For the model of part (c), find the limiting amount of salt in the tank as $t \to +\infty$, and interpret the result physically. (3 marks)

EXERCISE 24 **GCE-style essay question: beyond separability**

(14 marks) Consider the differential equation

$$\frac{dy}{dx} = \frac{x^2 + y^2}{xy}, \qquad x > 0,\ y > 0.$$

(a) Show that this equation is not separable, i.e. that its right-hand side cannot be written as a product $f(x)\,g(y)$. (3 marks)

(b) Using the substitution $y = vx$, where v is a function of x, show that

$$x\frac{dv}{dx} = \frac{1}{v}.$$

(4 marks)

(c) Solve the equation obtained in (b) and deduce the general solution of the original equation in the form $y^2 = 2x^2 \ln x + Cx^2$. (4 marks)

(d) Find the particular solution satisfying $y(1) = 2$ and state its domain. (3 marks)

(Bonus, 2 marks: verify by direct differentiation that your solution of part (c) satisfies the original equation.)

9 SOLUTIONS TO THE EXERCISES

SOLUTION **Exercise 1 — Multiple choice: identifying a separable equation**

The correct answer is (b).

1. $\frac{dy}{dx} = x + y$: a sum of a function of x and a function of y; it cannot be factored into $f(x)g(y)$. Not separable.
2. $\frac{dy}{dx} = e^{x+y} = e^x \cdot e^y$: a product of a function of x alone and a function of y alone. Separable.
3. $\frac{dy}{dx} = \sin(x + y)$: the argument $x + y$ cannot be split into a product $f(x)g(y)$ (the identity $\sin(x + y) = \sin x \cos y + \cos x \sin y$ gives a sum, not a product). Not separable.
4. $\frac{dy}{dx} = xy+x+y+1$: although this factors as $(x+1)(y+1)$, note carefully: $xy+x+y+1 = (x+1)(y+1)$, which is a product of a function of x and a function of y! So (d) is also separable. In a typical multiple-choice setting where only one answer is expected, (b) is the intended answer; however, a careful student should observe that (d) factorises as $(x+1)(y+1)$ and is therefore separable too. In an examination, mention both and justify.

Answer: (b) (with the remark that (d) is also separable since $xy + x + y + 1 = (x + 1)(y + 1)$).

SOLUTION **Exercise 2 — True or false?**

1. True. By definition, $\frac{dy}{dx} = f(x)g(y)$ separates as $\frac{dy}{g(y)} = f(x)\,dx$ (for $g(y) \neq 0$), and both sides can be integrated.
2. True. Separating: $\frac{dy}{y} = 3\,dx$, so $\ln|y| = 3x+C_1$, hence $|y| = e^{C_1}e^{3x}$, i.e. $y = Ce^{3x}$ with $C \in \mathbb{R}$ (including $C = 0$, the constant solution). Check: $y' = 3Ce^{3x} = 3y$. ✓
3. True. If $y = 2e^{x^2}$, then $\frac{dy}{dx} = 2e^{x^2}\cdot 2x = 2x\left(2e^{x^2}\right) = 2xy$, and $y(0) = 2e^0 = 2$. Both conditions hold. ✓
4. True. Separating $\frac{dy}{dx} = y^2$ gives $\frac{dy}{y^2} = dx$, valid only for $y \neq 0$. Yet $y = 0$ clearly satisfies the equation ($0' = 0 = 0^2$). This constant solution is lost when dividing by y^2; it must be checked and quoted separately.

SOLUTION **Exercise 3 — Definitions**

1. A **first-order differential equation** is an equation involving an unknown function y of one variable x, its derivative $\frac{dy}{dx}$, and possibly x itself; it contains no derivative of order higher than one. General form: $F\left(x, y, \frac{dy}{dx}\right) = 0$. Example: $\frac{dy}{dx} = 3x^2y$.

2. The **general solution** is the family of all solutions, expressed with one arbitrary constant C (one constant because the equation is of first order, so one integration is needed). A **particular solution** is obtained by assigning a specific value to C, usually determined by an initial condition. Example: $y' = 2x$ has general solution $y = x^2 + C$; with $y(0) = 3$, the particular solution is $y = x^2 + 3$.

3. An **initial condition** is a requirement of the form $y(x_0) = y_0$, specifying the value of the solution at one point. A first-order equation involves exactly one integration, producing exactly one arbitrary constant; one initial condition gives exactly one equation to determine that constant, hence a unique particular solution (geometrically: one point selects one curve from the family of solution curves).

SOLUTION **Exercise 4 — Direct calculations**

1. $\frac{dy}{dx} = \frac{x}{y}$. Separate: $y\,dy = x\,dx$. Integrate: $\frac{y^2}{2} = \frac{x^2}{2} + C_1$, i.e.

$$y^2 = x^2 + C \qquad (C \in \mathbb{R}).$$

(Solution curves are hyperbolas; for $C = 0$, the lines $y = \pm x$.)

2. $\frac{dy}{dx} = e^{2x}y$. Separate ($y \neq 0$): $\frac{dy}{y} = e^{2x}\,dx$. Integrate: $\ln|y| = \frac{1}{2}e^{2x} + C_1$, hence

$$y = Ce^{\frac{1}{2}e^{2x}} \qquad (C \in \mathbb{R},\ C = 0 \text{ included}).$$

3. $\frac{dy}{dx} = \frac{1}{x^2y}$ with $x > 0$, $y > 0$. Separate: $y\,dy = \frac{dx}{x^2}$. Integrate: $\frac{y^2}{2} = -\frac{1}{x} + C_1$, hence

$$y^2 = C - \frac{2}{x}, \qquad \text{i.e. } y = \sqrt{C - \frac{2}{x}} \quad \text{(positive root since } y > 0).$$

4. $x\frac{dy}{dx} = y + 1$ with $y > -1$. Separate: $\frac{dy}{y+1} = \frac{dx}{x}$. Integrate: $\ln(y+1) = \ln|x| + C_1$, so $y + 1 = Ax$ with $A > 0$ if $x > 0$; in general the family of solutions is

$$y = Cx - 1 \qquad (C \in \mathbb{R}).$$

The constant solution $y = -1$ corresponds to $C = 0$. The condition $y > -1$ excludes $C = 0$ and restricts the domain according to the sign of Cx. Check: $x\frac{dy}{dx} = xC = y + 1$. ✓

SOLUTION **Exercise 5 — An initial-value problem with an explicit solution**

Separation. $\frac{dy}{dx} = \frac{3x^2}{2y}$ gives $2y\,dy = 3x^2\,dx$.
Integration. $y^2 = x^3 + C$.
Initial condition. $y(1) = 2$: $4 = 1 + C$, so $C = 3$. Hence $y^2 = x^3 + 3$.
Sign of the root. $y = \pm\sqrt{x^3 + 3}$. Since $y(1) = 2 > 0$, we must take the positive root:

$$\boxed{y = \sqrt{x^3 + 3}}.$$

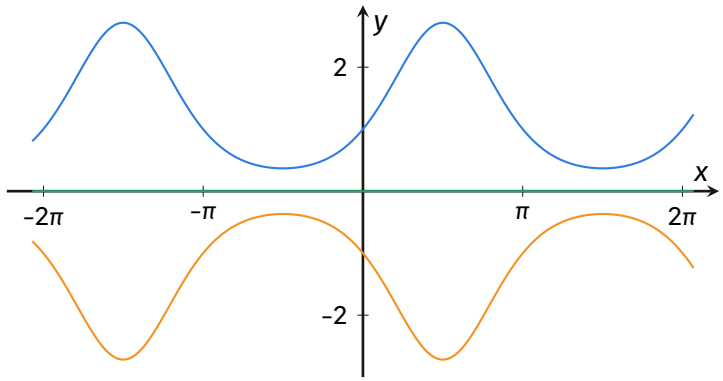
Domain of validity. We need $x^3 + 3 > 0$ (strictly, since y appears in the denominator of the equation, $y \neq 0$), i.e. $x > -\sqrt[3]{3}$. The solution is valid on $\left]-\sqrt[3]{3}, +\infty\right[$, the largest interval containing $x_0 = 1$ on which the solution is defined and differentiable.

SOLUTION **Exercise 6 — Exponential growth of a bacterial culture**

1. Separate: $\frac{dN}{N} = 0.05\,dt$, so $\ln N = 0.05t + C_1$ ($N > 0$), i.e. $N = Ae^{0.05t}$. With $N(0) = 500$: $A = 500$. Hence
$$\boxed{N(t) = 500\,e^{0.05t}} \quad \text{(in thousands)}.$$
2. Tripling: $N(t) = 3N(0)$, i.e. $e^{0.05t} = 3$, so $t = \frac{\ln 3}{0.05} = 20 \ln 3 \approx 20 \times 1.0986 \approx 21.97$. To the nearest hour, the population triples after $\boxed{22 \text{ hours}}$.
3. In general, with $N(0) = N_0$: $N(t) = N_0 e^{0.05t}$. Tripling means $N_0 e^{0.05t} = 3N_0$; dividing by N_0 gives $e^{0.05t} = 3$, so $t = 20 \ln 3$ independently of N_0. This is a characteristic feature of exponential growth: the time to multiply by any fixed factor is constant.

SOLUTION **Exercise 7 — Trigonometric and logarithmic separation**

1. $\frac{dy}{dx} = y \cos x$. Separate ($y \neq 0$): $\frac{dy}{y} = \cos x\,dx$, so $\ln|y| = \sin x + C_1$, hence
$$y = Ce^{\sin x} \qquad (C \in \mathbb{R}).$$
Through $(0, 1)$: $1 = Ce^0 \Rightarrow C = 1$, curve $y = e^{\sin x}$. Through $(0, -1)$: $C = -1$, curve $y = -e^{\sin x}$. Through $(0, 0)$: $C = 0$, the constant solution $y = 0$ (the x-axis).



Curves $y = e^{\sin x}$ (blue), $y = -e^{\sin x}$ (orange), $y = 0$ (green)

Note that solution curves never cross: $e^{\sin x} > 0$ always, so the sign of y is fixed by the initial condition.

2. $\dfrac{dy}{dx} = \dfrac{y \ln x}{x}$, $x > 0$. Separate: $\dfrac{dy}{y} = \dfrac{\ln x}{x}\,dx$. Since $\displaystyle\int \frac{\ln x}{x}\,dx = \frac{(\ln x)^2}{2}$ (substitution $u = \ln x$), we get $\ln|y| = \dfrac{(\ln x)^2}{2} + C_1$, i.e. $y = Ce^{(\ln x)^2/2}$. With $y(1) = 3$: $\ln 1 = 0$, so $C = 3$. Hence

$$\boxed{y = 3e^{\frac{(\ln x)^2}{2}}}, \qquad x > 0.$$

SOLUTION **Exercise 8 — Newton's law of cooling**

1. Let $\theta = T - 20$. Then $\dfrac{d\theta}{dt} = \dfrac{dT}{dt} = -k\theta$. Separating: $\dfrac{d\theta}{\theta} = -k\,dt$, so $\ln\theta = -kt + C_1$, i.e. $\theta = Ae^{-kt}$. Hence

$$T(t) = 20 + Ae^{-kt}.$$

$T(0) = 90$: $A = 70$. So $T(t) = 20 + 70e^{-kt}$. $T(5) = 70$: $70 = 20 + 70e^{-5k}$, so $e^{-5k} = \dfrac{50}{70} = \dfrac{5}{7}$, giving

$$k = \frac{1}{5}\ln\frac{7}{5} \approx \frac{0.3365}{5} \approx 0.0673\ \text{min}^{-1}.$$

2. $T(15) = 20 + 70e^{-15k} = 20 + 70\left(e^{-5k}\right)^3 = 20 + 70\left(\dfrac{5}{7}\right)^3 = 20 + 70 \times \dfrac{125}{343} \approx 20 + 25.5 \approx \boxed{46^\circ\text{C}}$.

3. As $t \to +\infty$, $e^{-kt} \to 0$ (since $k > 0$), so $T(t) \to 20^\circ$C. The tea tends to the room temperature, which is exactly what experience predicts: cooling never goes below the ambient temperature in this model.

SOLUTION **Exercise 9 — GCE-style: curve through a point**

(a) The gradient at (x, y) is $\dfrac{dy}{dx}$, so

$$\frac{dy}{dx} = \frac{2y}{x} = \underbrace{\frac{2}{x}}_{f(x)} \cdot \underbrace{y}_{g(y)},$$

a product of a function of x alone and a function of y alone: the equation is separable. [M1: equation; A1: justification]

(b) For $y \neq 0$: $\dfrac{dy}{y} = \dfrac{2}{x}\,dx$. Integrating: $\ln|y| = 2\ln|x| + C_1 = \ln x^2 + C_1$, so $|y| = e^{C_1}x^2$, i.e.

$$y = Cx^2 \qquad (C \in \mathbb{R},\ C = 0 \text{ included}).$$

[M1: separation; M1: correct integrals; A1: ln form; A1: explicit form $y = Cx^2$]

(c) Through $(1, 2)$: $2 = C \cdot 1$, so $C = 2$ and $\boxed{y = 2x^2}$. Since the original equation requires $x \neq 0$ and the curve passes through $x = 1$, the solution is valid for $x > 0$. [M1: substitution of point; A1: $y = 2x^2$ with domain]

(d) At $x = 2$: $y = 2(2)^2 = 8$. Gradient: $\dfrac{dy}{dx} = \dfrac{2y}{x} = \dfrac{16}{2} = 8$. Tangent:

$$y - 8 = 8(x - 2) \quad\Longrightarrow\quad \boxed{y = 8x - 8}.$$

[M1: point and gradient; A1: equation of tangent]

Examiner's expectations: clear separation with dy and dx on opposite sides; constant of integration shown; domain stated (a mark is often lost for omitting $x > 0$).

SOLUTION **Exercise 10 — GCE-style: a mixing problem**

(a) Volume in per minute: 3 L; out: 2 L; net gain 1 L/min. Starting from 100 L: $V(t) = 100 + t$ litres. [B1]

(b) Salt enters at rate $0.2 \times 3 = 0.6$ kg/min. The mixture leaves at 2 L/min with concentration $\dfrac{S}{V} = \dfrac{S}{100 + t}$ kg/L, so salt leaves at rate $\dfrac{2S}{100 + t}$ kg/min. Therefore

$$\frac{dS}{dt} = 0.6 - \frac{2S}{100 + t}.$$

The right-hand side is a difference of a constant and a term involving both t and S; it cannot be written as a product $f(t)\,g(S)$, so the equation is not separable as it stands. [M1: inflow rate; M1: outflow rate; A1: non-separability justified]

(c) Outflow now 3 L/min, volume constant at 100 L. Salt leaves at $3 \times \dfrac{S}{100} = \dfrac{3S}{100}$ kg/min, so

$$\frac{dS}{dt} = 0.6 - \frac{3S}{100} = \frac{60 - 3S}{100} = \underbrace{\frac{3}{100}}_{f(t)\text{ (constant)}} \cdot \underbrace{(20 - S)}_{g(S)},$$

which is separable. Separating (for $S \neq 20$):

$$\frac{dS}{20-S} = \frac{3}{100}\,dt \implies -\ln|20-S| = \frac{3t}{100} + C_1 \implies 20 - S = Ae^{-3t/100}.$$

$S(0) = 0$: $A = 20$. Hence

$$\boxed{S(t) = 20\left(1 - e^{-0.03t}\right)}.$$

[M1: new equation; A1: separable shown; M1: separation and integration; M1: exponential form; A1: constant from $S(0) = 0$]

(d) As $t \to +\infty$, $e^{-0.03t} \to 0$, so $S(t) \to 20$ kg. Interpretation: the salt content approaches the equilibrium where outflow removes salt as fast as inflow supplies it; the concentration tends to $\frac{20}{100} = 0.2\,\text{kg/L}$, exactly the concentration of the incoming brine — the tank's content gradually becomes indistinguishable from the inflow. [M1: limit; A1: value 20 kg; A1: physical interpretation]

Examiner's expectations: units used consistently; clear distinction between the non-separable model (b) and the separable one (c); equilibrium interpreted, not merely computed.

SOLUTION **Exercise 11 — GCE-style essay: beyond separability**

(a) Suppose $\frac{x^2+y^2}{xy} = f(x)g(y)$ for all $x, y > 0$. Note $\frac{x^2+y^2}{xy} = \frac{x}{y} + \frac{y}{x}$. Setting $x = y = 1$ gives $f(1)g(1) = 2$; setting $x = 1, y = 2$ and $x = 2, y = 1$ both give $\frac{5}{2}$, so $f(1)g(2) = f(2)g(1)$. Setting $x = y = 2$ gives $f(2)g(2) = 2$. Then

$$\big(f(1)g(2)\big)\big(f(2)g(1)\big) = \tfrac{25}{4}, \qquad \big(f(1)g(1)\big)\big(f(2)g(2)\big) = 4,$$

but the two left-hand sides are equal (both equal $f(1)f(2)g(1)g(2)$), while $\frac{25}{4} \neq 4$: contradiction. Hence no such factorisation exists and the equation is not separable. [M1: assumption; M1: test values; A1: contradiction]

(b) Let $y = vx$. Then $\frac{dy}{dx} = v + x\frac{dv}{dx}$. Also

$$\frac{x^2+y^2}{xy} = \frac{x^2+v^2x^2}{x\cdot vx} = \frac{1+v^2}{v}.$$

Therefore

$$v + x\frac{dv}{dx} = \frac{1+v^2}{v} = v + \frac{1}{v} \implies \boxed{x\frac{dv}{dx} = \frac{1}{v}}.$$

[M1: derivative of vx; M1: substitution; A1: simplification of RHS; A1: result]

(c) The new equation is separable: $v\,dv = \frac{dx}{x}$. Integrating:

$$\frac{v^2}{2} = \ln x + C_1 \quad (x > 0), \qquad \text{so } v^2 = 2\ln x + C.$$

Since $v = \dfrac{y}{x}$, $\dfrac{y^2}{x^2} = 2\ln x + C$, hence

$$\boxed{y^2 = 2x^2 \ln x + Cx^2}.$$

[M1: separation; M1: integration; A1: $v^2 = 2\ln x + C$; A1: back-substitution]
(d) $y(1) = 2$: $4 = 0 + C$, so $C = 4$ and

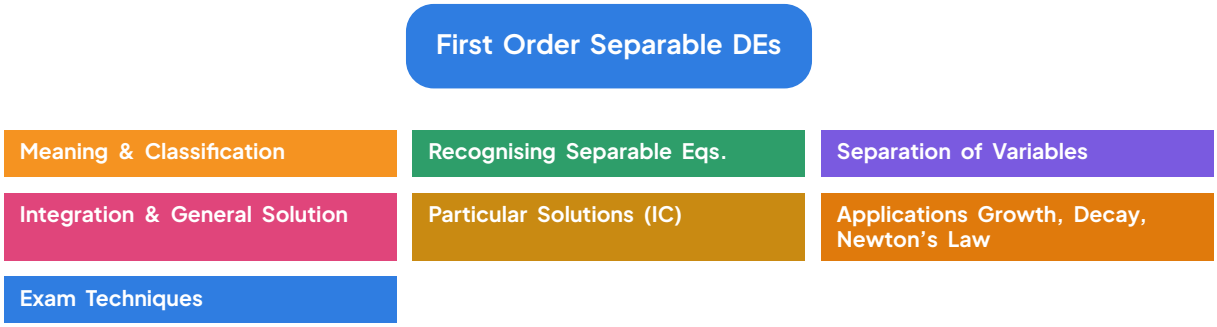
$$y^2 = 2x^2 \ln x + 4x^2 = 2x^2(\ln x + 2), \qquad y = x\sqrt{2(\ln x + 2)}$$

(positive root since $y > 0$). Domain: $\ln x + 2 > 0$, i.e. $x > e^{-2}$; the particular solution is valid on $]e^{-2}, +\infty[$. [M1: constant; A1: explicit y; A1: domain]
Bonus verification. From $y^2 = 2x^2 \ln x + Cx^2$, differentiate: $2y\dfrac{dy}{dx} = 4x\ln x + 2x + 2Cx$. Divide by $2y$:

$$\frac{dy}{dx} = \frac{2x\ln x + x + Cx}{y} = \frac{x(2\ln x + 1 + C)}{y}.$$

Now $\dfrac{x^2 + y^2}{xy} = \dfrac{x^2 + 2x^2\ln x + Cx^2}{xy} = \dfrac{x(1 + 2\ln x + C)}{y}$: identical. ✓
Examiner's expectations: rigorous contradiction argument in (a) (a mere "it looks like a sum" earns no marks); correct use of the product rule in (b); the back-substitution $v = y/x$ explicitly written in (c); domain in (d).

10 SUMMARY SHEET



Mind map of Chapter PMS05: First Order Differential Equations With Separable Variables

FICHE BILAN **L'essentiel à connaître pour l'examen**

Key Definitions

- **Differential Equation (DE)**: an equation involving an unknown function and its derivatives.
- **Order**: the highest derivative present (here first order).
- **Degree**: the power of the highest derivative after clearing fractions and radicals.
- **Separable DE**: can be written as $\frac{dy}{dx} = g(x)h(y)$ or $M(x)dx + N(y)dy = 0$.
- **General Solution**: a family of solutions containing an arbitrary constant C.
- **Particular Solution**: a specific solution obtained by imposing an **initial condition** $y(x_0) = y_0$.

Fundamental Formula / Law

If $\frac{dy}{dx} = g(x)h(y)$ with $h(y) \neq 0$, then

$$\int \frac{1}{h(y)}\,dy = \int g(x)\,dx + C.$$

After integration, solve for y explicitly if possible; otherwise keep the implicit relation.

Step-by-Step Method

1. Identify whether the DE is separable: write it in the form $\frac{dy}{dx} = g(x)h(y)$.
2. Separate variables: bring all y terms (including dy) to one side and all x terms (including dx) to the other.
3. Integrate both sides: $\int \frac{dy}{h(y)} = \int g(x)\,dx + C$. Remember the constant of integration on one side only.

4. Simplify the result: combine logarithms, exponentiate, or solve algebraically for y.
5. Apply initial condition (if given): substitute x_0, y_0 to find C, then write the particular solution.
6. Check by differentiating your solution and verifying it satisfies the original DE.

Common Mistakes (Traps)

- Forgetting the constant C after integration.
- Dividing by $h(y)$ without considering the case $h(y) = 0$ (which may give singular solutions).
- Incorrect separation: e.g. treating $\frac{dy}{dx} = x + y$ as separable (it is not).
- Misplacing the constant when exponentiating: $e^{\ln|y|+C} = e^C|y|$, not $|y| + C$.
- Not stating the domain of the solution, especially when logarithms or square roots appear.
- Failing to simplify the final answer (e.g. leaving $\ln|y| = x^2 + C$ instead of $y = \pm e^C e^{x^2}$).

GCE Exam Tips

- Show every integration step; examiners award marks for correct separation and integration even if the final algebra has a minor slip.
- When an initial condition is given, substitute before solving for y explicitly if it avoids messy algebra.
- If the DE models a real situation (population growth, radioactive decay, cooling), define variables clearly and state the model equation (e.g. $\frac{dP}{dt} = kP$).
- Practise past GCE questions: typical tasks are "Find the general solution", "Find the particular solution given $y(0) = 2$", and "Interpret the constant k in the context".
- Use the variation table (sign of derivative) to discuss the behaviour of solutions when asked to sketch or comment on long-term trends.

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